

Experimental Analysis: Privacy-Preserving Few-Shot UAV Identification

Baseline Study — Synthetic Dataset, Prototypical Networks, MobileNetV3-Small
KAI Vision Platform — TFG — University of Alicante / Embention

1. Introduction and Motivation

The central goal of this work is to enable a ground-based or airborne sensor platform to identify a specific unmanned aerial vehicle (UAV) from a small number of reference photographs supplied by a third-party client, without those photographs ever leaving the client's system. This requirement, which we term the privacy-preserving few-shot UAV identification problem, combines two constraints that are rarely addressed simultaneously in the literature: the privacy constraint eliminates any server-side processing of client imagery, and the few-shot constraint eliminates retraining the model on new targets.

Existing counter-UAV systems typically address detection (is there a UAV present?) rather than identification (is this the specific UAV we are looking for?). The UAV-ReID benchmark established by Organisciak et al. (2021) represents the only published work on UAV re-identification as of the time of writing, and its Big-to-Small sub-challenge — matching a large close-range image to a small distant crop — achieved only 46.5% mAP with the best evaluated architecture. This result quantifies the difficulty of the cross-scale matching problem inherent in our scenario, where a client supplies high-quality reference photographs but the device observes the target at distances of 50 to 100 metres.

The approach developed in this thesis draws on the metric learning literature, specifically Prototypical Networks (Snell, Swersky & Zemel 2017), which learn a general-purpose embedding space rather than a fixed set of class labels. At deployment, enrolling a new target requires only a forward pass through the encoder — no retraining, no server communication — making the architecture compatible with the strict privacy requirements of defence clients.

The experimental work proceeded iteratively. Each stage was designed to answer a specific question before the next could begin: first validating that the metric learning concept works on synthetic data, then identifying and solving the cross-domain gap between enrollment photographs and operational footage, then quantifying the contribution of real imagery, and finally aligning the training pipeline with the exact input format produced by the YOLOX detector at runtime. This progression is documented in full in Section 6, including the intermediate results that motivated each decision. Section 6.16 provides an explicit account of why each stage was necessary rather than optional.

2. System Architecture

2.1 Pipeline Overview

The identification system is structured as a two-stage pipeline. The first stage is a generic detection model — a YOLOX-tiny variant trained by Embention on the Airborne Object Tracking dataset for the classes airplane, helicopter, bird, and undetermined — whose sole responsibility is to produce tight bounding-box crops of all flying objects in the camera frame. The second stage, developed in this thesis, is the identification model, which receives those crops and determines whether any of them matches the enrolled target.

It is important to note that the YOLOX detection stage was not used in this baseline study. The Blender renders are UAV-centric by construction: each image contains only the UAV against a sky background, with no need for a detection step. The full 640x640 render is passed directly to the encoder. YOLOX enters the pipeline only when moving to real footage, where the UAV is a small object embedded in a larger scene that must first be localised and cropped. The two-stage architecture described above is the planned production design; this study validates the identification stage in isolation.

2.2 Encoder Architecture

The encoder backbone is MobileNetV3-Small (Howard et al. 2019), a depthwise-separable convolutional architecture designed for embedded inference. With approximately 2.5 million parameters and a final feature map of dimensionality 576, it provides a strong accuracy-efficiency trade-off validated on ImageNet. The backbone is pretrained on ImageNet-1K and fine-tuned episodically; the pretrained initialisation is essential given the small number of UAV training identities available (16 in this study).

On top of the backbone, a compact projection head maps the 576-dimensional pooled features to a 128-dimensional embedding space. The head consists of a single linear layer followed by LayerNorm and L2 normalisation. The L2 normalisation step places all embeddings on the unit hypersphere, making cosine similarity and Euclidean distance rank-equivalent for all subsequent comparisons. This property is important for the privacy architecture: because all embeddings lie on the unit sphere with bounded, comparable magnitudes, the gallery file (gallery.npy) can be calibrated with a single global distance threshold rather than per-identity thresholds.

The combined encoder maps any input image of arbitrary UAV class to a 128-dimensional L2-normalised vector. Identical airframes produce embeddings that are close in this space; different airframes produce embeddings that are far apart. This property is learned during training and generalises to airframes never seen during training — the defining characteristic that makes the approach suitable for open-set identification of novel client targets.

2.3 Embedded Deployment Split

The TI TDA4VM and AM69A system-on-chip used by the KAI platform accelerates deep learning through the TIDL graph compiler. TIDL does not support the LayerNorm operator used in the projection head. The model is therefore exported in two ONNX artefacts: `backbone_tidl.onnx`, containing the MobileNetV3-Small features and global average pooling, runs entirely on the C7x/MMA deep-learning accelerator and accounts for more than 95% of the compute. `projection_head.onnx`, containing the Linear layer, LayerNorm, and L2 normalisation, runs on the ARM Cortex-A cores via ONNX Runtime at negligible cost, since it consists of a single small matrix multiplication followed by elementwise operations. This split eliminates the LayerNorm incompatibility while preserving exact numerical behaviour and allowing INT8 quantisation of the backbone through TIDL's calibration pipeline.

3. Privacy Architecture

The governing requirement is that client images must never reach Embention or any external server. The design, referred to as Option A, satisfies this by construction rather than by policy. The client installs an enrollment application on their own system. The application ships the encoder in ONNX format and executes it locally, producing a `gallery.npy` file containing the L2-normalised mean embedding of the N enrolled reference photographs. This file — approximately 0.5 KB for a 128-dimensional vector — is the only artefact that ever leaves the client's premises and is transferred to the KAI device.

The irreversibility of this transfer is a critical security property. An embedding is a lossy many-to-one compression: the original image cannot be reconstructed from a 128-dimensional vector. At most, an attacker in possession of `gallery.npy` could verify whether a specific image they already hold produces a similar embedding — a comparison oracle attack that requires the attacker to already possess candidate images and yields only a similarity verdict, not a reconstruction. A formal analysis of this property, including evaluation against published embedding-inversion attacks (Mai et al. 2019, applied to the face domain), is planned for the later phases of this work.

Compared with alternative privacy strategies considered during the feasibility study — federated learning (leaks gradients, complex, unnecessary given that the model trains on public data), secure enclaves (hardware-dependent, premature for the current phase), and on-device training (requires the client to own identical hardware) — Option A is the only approach that is simultaneously simple, auditable, and provably compliant with the data-residency constraint without requiring any client-side training infrastructure.

4. Dataset Construction

4.1 The Fixed-Wing Identity Scarcity Problem

Public UAV datasets used in the counter-UAV literature — including the Anti-UAV challenge dataset, VisioDECT, DUT Anti-UAV, and MAV-VID — consist almost entirely of multirotor

platforms (DJI Phantom, Mavic, Inspire series). The operational target class for this project is a fixed-wing airframe of approximately 2 metres wingspan, which has a fundamentally different silhouette: swept or straight wings, tail configuration, absence of rotor arms. No public dataset provides identity-labelled fixed-wing UAV crops at the operational pixel envelope. This scarcity is the primary motivation for the synthetic rendering pipeline described in Section 4.2.

4.2 Synthetic Rendering Pipeline

Three-dimensional models of fixed-wing UAVs were obtained from Sketchfab under Creative Commons Attribution licences, including models of the MQ-1 Predator, MQ-9 Reaper, Orion UAV, and several additional fixed-wing platforms. Hard-negative identities — classes the system must reject at deployment — were obtained from models of quadrotor UAVs, helicopters, and birds. Each model was imported into Blender 4.x and rendered using the Cycles path-tracing engine.

The rendering pipeline was automated using a custom Python script executed headlessly from the command line. Each model was normalised to unit scale (longest axis = 1.0 Blender unit) and centred at the origin. A camera was placed at 36 Blender units distance with an 85mm focal length, producing a target angular size of approximately 40 to 60 pixels wide in a 640×640 render — consistent with the 46 to 143 pixel operational envelope derived from the KAI sensor geometry (1920×1080, 30° to 45° HFOV, 50 to 100 metre engagement range). Five distinct sky conditions were used: three Nishita physically-based sky textures at varying solar elevations (morning, noon, and dusk) and two flat overcast colours. The camera orbited each model at 6 azimuth angles (every 60°) and 3 elevation angles (−10°, 0°, 25°), producing 54 renders per model ($6 \times 3 \times 3$ sky conditions).

Separate enrollment renders were produced with the camera at 3.8 Blender units distance, placing the UAV at approximately 380 pixels wide — simulating the close-range reference photographs a client would supply. These enrollment renders are used exclusively for evaluation of the Q1 and Q3 experiments; they are not seen during training.

4.3 Dataset Statistics and Split

The final dataset comprises 22 distinct identities across three splits. The train split contains 16 identities (9 fixed-wing UAV models and 7 hard negatives) with 54 images each, totalling 864 images. The val split contains 6 unseen fixed-wing UAV identities with 54 images each, totalling 324 images. The enrollment split contains 12 positive identities (covering both train and val) with 54 close-range renders each, totalling 648 images. The identity-disjoint constraint between train and val is enforced by construction: the batch rendering pipeline uses distinct model subfolders for each split, and an automated audit script verifies zero overlap before any training run.

Hard-negative identities are placed exclusively in the train split and are never present in val or enrollment. This reflects their role in training: they provide contrast examples that teach the embedding space to push non-UAV flying objects away from UAV embeddings, which is

essential for the open-set deployment scenario where the system must also reject birds, conventional aircraft, and multirotors when a fixed-wing UAV is enrolled.

5. Training Methodology

5.1 Episodic Prototypical Training

The encoder is trained using the episodic prototypical learning scheme of Snell et al. (2017). In each training episode, N identities are sampled from the training set (N -way, $N = 10$ in this study), and for each identity K support images and Q query images are sampled (K -shot, $Q = 5$). The support images are embedded and their mean per-identity embedding is taken as the class prototype. Query embeddings are then compared against all prototypes using squared Euclidean distance, and the training objective is a softmax cross-entropy loss over the resulting logits. The choice of squared Euclidean distance over cosine similarity is motivated by the theoretical result of Snell et al. that squared Euclidean distance is a Bregman divergence, making the class mean the optimal prototype under this metric. On the unit sphere (where embeddings reside after L2 normalisation), the two metrics are rank-equivalent, but the Euclidean formulation is kept for theoretical consistency and to enable a meaningful unnormalised ablation.

5.2 Training Configuration

Several departures from the original Snell et al. protocol are deliberately adopted, motivated by the differences between the standard few-shot classification benchmarks (Omniglot, miniImageNet) and this deployment context.

First, the train-way (10) is set higher than the test-way (5), following the recommendation of Snell et al. that harder training episodes improve generalisation. With 16 training identities available, 10-way episodes are the highest feasible setting without degenerate episode sampling.

Second, the shot count is randomised per episode from the set $\{1, 3, 5\}$, producing a shot-robust model that handles any enrollment size. The original paper recommends matching train and test shot, which is valid when the test-time shot count is fixed by protocol. In our deployment scenario, the test-time shot count is the number of images the client uploads — the very quantity under study in the Q1 experiment — so a single model that generalises across shot counts is preferable to separate checkpoints per shot value.

Third, the backbone uses a lower learning rate (10^{-4}) than the projection head (10^{-3}) for the first two epochs, during which the backbone is frozen entirely. This prevents the pretrained ImageNet features from being destroyed before the projection head has learned a meaningful initial metric. After epoch 2, all parameters are jointly fine-tuned with AdamW and mild weight decay (10^{-4}), using cosine annealing over the full 30-epoch training run. The original paper used Adam without weight decay on from-scratch shallow networks; with a

2.5M-parameter pretrained backbone and only 864 training images, some weight decay is necessary to prevent overfitting.

6. Experimental Results

6.1 Training Dynamics

Training ran for 30 epochs on an NVIDIA RTX 3090 (approximately 70 seconds per epoch). Training accuracy increased consistently from 0.183 at epoch 1 to 0.832 at epoch 30, reflecting effective metric learning on the training identities. Validation accuracy followed training accuracy closely for the first 17 epochs, reaching a peak of 0.650 at epoch 17, after which it plateaued and oscillated between 0.60 and 0.66. The best checkpoint (val acc = 0.657, epoch 24) was retained for all subsequent evaluation.

The gap between training accuracy (0.832) and validation accuracy (0.657) from epoch 18 onwards indicates a degree of overfitting to the synthetic training domain. The training set consists entirely of Blender renders with clean sky backgrounds and no domain randomisation of surface textures or material properties. The encoder learns rendering-specific features — consistent illumination, crisp silhouettes, absence of motion blur — that do not fully generalise to real operational imagery. This is an expected limitation of the purely synthetic first study and motivates the inclusion of real-imagery augmentation in subsequent experiments.

It is important to note that the validation accuracy reported during training (5-way episodic, 100 episodes) carries significant variance due to the small val set (6 identities). The oscillations observed after epoch 18 are consistent with this variance and should not be interpreted as instability in the training dynamics. The open-set ROC-AUC reported in Section 6.2, computed over the full val set in a deterministic evaluation protocol, is a more reliable indicator of model quality.

6.2 Open-Set Identification Performance

The primary evaluation metric for the deployment task is the ROC-AUC for target-versus-impostor discrimination in an open-set protocol. For each val identity, K images are randomly sampled as the enrollment set, and the mean embedding of those K images forms the prototype. All remaining images of that identity are scored as genuine queries, and images of all other val identities (3 images each) are scored as impostors. The cosine similarity score (equivalent to negative squared Euclidean distance on the unit sphere) determines the ranking.

The following table reports ROC-AUC and TPR at fixed FPR operating points for mean aggregation (mean prototype) and attention-weighted aggregation (Approach 2) at $k_shot = 5$:

Aggregation	ROC-AUC	Genuine mean	Impostor mean	TPR@FPR=1%	TPR@FPR=5%	TPR@FPR=10%
Mean	0.828	0.697	0.168	0.133	0.328	0.433
Attention	0.812	0.858	0.467	0.206	0.328	0.489

Mean aggregation achieves a ROC-AUC of 0.828 at $k_{\text{shot}} = 5$, which represents the primary go/no-go metric for this phase of the study. A ROC-AUC of 0.828 on identities never seen during training, using a purely synthetic training dataset with no real UAV imagery, demonstrates that the prototypical network approach is viable for this task.

Attention-weighted aggregation achieves a lower ROC-AUC (0.812) despite producing a substantially higher mean genuine score (0.858 vs 0.697). The explanation lies in the impostor score distribution: attention raises impostors from 0.168 to 0.467, nearly closing the genuine-impostor gap. The attention mechanism selects the gallery view most similar to the current query and weights the prototype accordingly. When the query is an impostor from a visually similar airframe class, this mechanism over-commits to a superficially similar enrolled angle and produces an inflated similarity score. With only 5 enrolled views, all of which are clean synthetic renders at uniformly spaced angles, the attention weight distribution is underdetermined and collapses to the single most similar view for any query. This finding is important: attention aggregation is theoretically motivated, but its benefit only materialises when the gallery is diverse enough (many viewpoints at varying distances and lighting) for the attention distribution to be informative. In the current dataset, the gallery diversity is insufficient, and mean aggregation is the more robust choice.

The TPR values at strict FPR operating points (1%, 5%, 10%) reveal the principal limitation of this baseline study. At FPR = 1%, mean aggregation achieves only TPR = 13.3%, meaning the threshold that allows one false alarm per 100 impostor observations also rejects 86.7% of genuine targets. This is operationally insufficient and is attributable to two causes: first, the genuine and impostor score distributions, while well-separated in mean (gap of 0.53), have overlapping tails produced by difficult query angles not well represented in the enrollment gallery; second, all data is synthetic, so the threshold calibrated on synthetic renders does not generalise to real operational noise, blur, and background clutter. Both causes are addressed in the subsequent experimental phases described in Section 8.

6.3 Q1: Effect of Enrollment Size

The Q1 experiment measures identification performance as a function of the number of enrollment images k_{shot} , providing the empirical basis for the client-facing enrollment specification. Results are reported for $k_{\text{shot}} \in \{1, 3, 5, 10, 20\}$ using mean aggregation:

k_{shot}	ROC-AUC	Genuine mean	Impostor mean	TPR@1%	TPR@5%	TPR@10%
1	0.802	0.559	0.065	0.200	0.228	0.411

3	0.805	0.641	0.137	0.122	0.239	0.411
5	0.828	0.697	0.168	0.133	0.328	0.433
10	0.861	0.734	0.160	0.256	0.328	0.494
20	0.852	0.714	0.157	0.322	0.400	0.456

ROC-AUC increases monotonically from $k=1$ (0.802) to $k=10$ (0.861) before declining slightly at $k=20$ (0.852). The peak at $k=10$ indicates that additional enrollment images beyond this point introduce redundant views that dilute rather than improve the mean prototype. The curve has not yet reached a clear asymptote at $k=5$, which is the assumed default enrollment budget — this confirms that requesting 10 images from the client is justified and measurably beneficial.

The improvement from $k=1$ to $k=3$ is minimal (+0.003 AUC), while the jump from $k=5$ to $k=10$ is the largest single step (+0.033). This non-linear pattern suggests that the first few enrollment images establish the coarse prototype location in embedding space, while additional images beyond $k=5$ provide increasingly useful coverage of different viewing angles that reduce intra-class variance in the prototype. The marginal value of each image diminishes significantly after $k=10$.

A notable anomaly appears in the TPR@1% column: $k=1$ (0.200) outperforms $k=3$ (0.122) and $k=5$ (0.133) despite having the lowest AUC. This is a threshold calibration artefact, not a genuine accuracy advantage. With a single poorly-positioned prototype ($k=1$), impostor scores are very low (mean 0.065) because no impostor happens to be close to a single enrolled view. The 1% FPR threshold therefore falls to a very low value, accidentally admitting more genuine queries. As k increases, the prototype stabilises in a more central location, pulling up impostor scores (0.137 at $k=3$, 0.168 at $k=5$) and raising the 1% FPR threshold, which then excludes more genuine tail samples. ROC-AUC, which integrates over all thresholds, correctly identifies $k=1$ as the worst performer. This illustrates why AUC is a more appropriate primary metric than TPR at a fixed threshold when the threshold calibration dataset is small.

The practical recommendation derived from the initial Q1 result is a minimum enrollment of 5 images and an optimal enrollment of 10. This recommendation was subsequently revised following the colour mild clean dataset experiments — see Section 6.7 for the updated finding.

6.4 Q3: Effect of Enrollment Degradation

The Q3 experiment isolates the contribution of the operational degradation augmentation pipeline. The DegradeToOperational transform, applied during training with probability $p=0.5$, randomly downscales each training image so its longer side falls within the 46-143px operational envelope derived from the KAI sensor geometry, optionally applies Gaussian

blur, and then upscales back to the model input size. The effect is to destroy fine texture and sharpness detail, forcing the encoder to learn UAV structure from silhouette and coarse shape features rather than render-specific surface appearance.

Two models were trained under identical conditions except for this flag: the baseline model (degrade_p=0.5, reported in Sections 6.2 and 6.3) and an ablation model (degrade_p=0.0, no degradation). Both were evaluated at k_shot=10 with mean aggregation. Results are as follows:

Config	Best Val Acc	Train/Val Gap	ROC-AUC	Genuine mean	Impostor mean	TPR @1%	TPR @10%
degrade_p=0.5 (baseline)	0.657	0.175	0.861	0.734	0.160	0.256	0.494
degrade_p=0.0 (ablation)	0.637	0.295	0.854	0.742	0.107	0.272	0.433

The degradation-trained model achieves higher ROC-AUC (+0.007) and substantially higher TPR@10% (+0.061). The most informative comparison is the train/val accuracy gap: 0.175 with degradation versus 0.295 without. The no-degradation model reaches a training accuracy of 0.932 against a val accuracy of only 0.637 — severe overfitting driven by the model memorising sharpness and render-specific texture features that are present in training images but absent at operational pixel sizes. Degradation acts as domain regularisation, closing the gap by forcing the encoder to rely on shape and structural features that survive downscaling.

A counterintuitive result in the table requires explanation: the no-degradation model achieves a lower impostor mean score (0.107 vs 0.160), which superficially suggests that it pushes impostors further away. This is a synthetic-domain artefact. The no-degradation model learns very clean, sharp class boundaries between high-resolution renders, producing large separations within the synthetic domain. However, these boundaries do not transfer to real operational imagery where both genuine targets and impostor aircraft appear blurry, hazy, and at reduced resolution. The degradation model's slightly higher impostor mean reflects a more conservative embedding space that has been regularised to work across the quality spectrum.

The current AUC difference (+0.007) understates the true benefit of degradation because the validation set consists of the same clean synthetic renders as the training set. Both models perform similarly in this setting. The decisive test will come when the evaluation moves to real operational crops at 46-143px, where the degradation-trained model will substantially outperform the clean-trained model because it has been exposed to downsampled, blurred UAV images throughout training. The Q3 experiment will therefore be re-run after real imagery is incorporated into the pipeline, at which point the gap is expected to widen significantly.

Q3 conclusion: the operational degradation pipeline provides consistent benefits across all metrics and should be enabled at `degrade_p=0.5` by default. It is the correct default configuration for all subsequent experiments.

Important addendum: this recommendation applies specifically when training on clean renders that have no built-in degradation. Once the rendering pipeline itself bakes degradation directly into every image — as done in the colour mild clean dataset (Section 6.6) — adding training-time degradation on top via `degrade_p` becomes redundant and counterproductive. The `degrade_p` transform was designed to compensate for the absence of render-time degradation, not to stack on top of it. Applying both simultaneously results in double-degradation: images that were already small, blurry, and noisy are further downsampled and blurred during training, pushing them below the threshold where any useful identity signal survives. All experiments using the colour mild clean dataset and its successors therefore use `degrade_p=0.0` as the correct setting, with degradation fully handled at render time.

6.5 Experiment: Aggressive Realistic Augmentation

6.5.1 Motivation and Setup

The previous experiments used clean Blender renders — images where the UAV is sharp, well-lit, perfectly centred, and free of any distortion. While the operational degradation pipeline (`degrade_p=0.5`) partially bridges the gap between those clean renders and real operational footage, it applies degradation at training time as a post-processing step on an otherwise clean image. A more realistic approach is to bake all degradation directly into the renders so that every training image already looks like what the device would actually see.

A new dataset was generated (`uav_dataset_realistic_50_100_final`) with the following modifications applied directly in the Blender rendering pipeline and as image-level post-processing:

- Random camera roll and bank angle, so the UAV is not always perfectly horizontal in the frame
- Random camera distance and focal length, constrained so operational images remain small and far away
- Random camera aim offset, so the UAV is not always perfectly centred — simulating imperfect detector crops
- Random exposure, gamma, sun strength, and multiple sky presets (noon, morning, dusk, overcast, white hazy)
- Image-level imperfections: motion blur, defocus blur, atmospheric haze, sensor noise, brightness and contrast variation, colour and saturation shift, JPEG compression artefacts, and low-resolution roundtrip downscaling

The dataset was also expanded: each identity received 144 images instead of 54, and the number of training identities increased from 16 to 17. Enrollment images received a milder version of these effects, reflecting the assumption that client reference photographs are taken under controlled conditions rather than captured mid-flight. Training was run for 30 epochs with identical hyperparameters to the baseline, except `degrade_p=0.0` (no additional degradation at training time, since it is now baked into the images).

6.5.2 Results

The results were a clear regression across all metrics:

Config	Val Acc	Train/Val Gap	ROC-AUC	Genuine	Impostor	TPR@10%
Clean renders + <code>degrade_p=0.5</code>	0.657	0.175	0.861	0.734	0.160	0.494
Realistic renders + <code>degrade_p=0.0</code>	0.460	0.148	0.734	0.662	0.303	0.160

ROC-AUC dropped from 0.861 to 0.734, and TPR@10% fell dramatically from 0.494 to 0.160. The train/val gap actually narrowed (0.148 vs 0.175), which at first glance seems positive, but this is misleading: both training and validation accuracy dropped together because the task became harder for every image in the dataset, not because generalisation improved.

6.5.3 Analysis: What Went Wrong

The key diagnostic is the impostor mean score, which nearly doubled from 0.160 to 0.303. The genuine mean barely changed (-0.072), but the impostor mean rose sharply (+0.143). This asymmetry points to a specific failure mode: embedding space collapse caused by over-aggressive augmentation.

To understand this intuitively: the identification system works by placing images of the same aircraft close together in a mathematical space, and images of different aircraft far apart. The model learns to do this by studying the structural differences between airframe shapes — wing geometry, tail configuration, fuselage profile. When the augmentation stack applies maximum motion blur, maximum sensor noise, maximum haze, and JPEG compression all at once to every image, the UAV silhouette becomes an indistinguishable blurry smudge. At that point, there is no structural information left to separate one airframe from another. All aircraft end up in the same region of the embedding space, genuine and impostor scores converge, and the gap that makes identification possible disappears.

An analogy: imagine trying to recognise a colleague from a photograph that has been heavily blurred, covered in static, darkened, and then saved at very low quality. You might still tell that there is a human in the image, but you cannot recognise who it is. The same

thing happened here — the model can still tell that the image contains a UAV, but it can no longer tell which UAV.

A secondary issue is that the model had not yet converged at epoch 30. The loss was still at 1.334 at the final epoch, compared to 1.014 for the baseline model, indicating that the harder augmentation requires significantly more training time. However, the impostor mean collapse is a structural problem that more training alone cannot fix — it requires addressing the augmentation severity.

6.5.4 What This Experiment Teaches Us

This experiment reveals an important principle: there is a sweet spot for augmentation severity. Too little augmentation (clean renders, no degradation) causes the model to overfit to synthetic rendering artefacts and fail on real imagery. Too much augmentation destroys the discriminative information that the model needs to tell aircraft apart. The goal is to find the level of degradation that looks like a real operational crop at 50-100 metres while still preserving enough structural detail for shape-based discrimination.

A practical rule of thumb emerges from comparing the two experiments: augmentation is beneficial as long as a human observer can still identify the type of aircraft in the image. If the image is so degraded that even a person cannot tell whether it is a fixed-wing UAV or a bird, it provides no useful training signal and actively hurts the embedding space. The next render iteration should implement a severity budget — a cap on the maximum combined degradation per image — so that the most extreme combinations of blur, noise, and haze cannot co-occur in the same image.

Two diagnostic experiments are planned to separate the evaluation protocol from the model quality. First, using clean enrollment images (client-style reference photographs) as the gallery and degraded val images as queries — which matches the real deployment scenario and may recover a significant portion of the lost AUC. Second, evaluating the realistic model on the clean synthetic val set from the earlier experiments, to determine whether the model learned a usable metric that simply needs cleaner inputs to express it. The results of these diagnostics will determine whether the current checkpoint is worth continuing to train or whether the render pipeline should be revised first.

6.6 Dataset v3: Colour Variation and Mild Realistic Augmentation

6.6.1 Changes from Previous Datasets

Following the failure of aggressive augmentation (Section 6.5), a third dataset was produced that targets the sweet spot between the two extremes: clean enough that the UAV silhouette is always recognisable, but varied enough that the model cannot memorise rendering artefacts. Three changes were made relative to the clean render dataset:

- Colour randomisation: the surface colour of every UAV is randomised per render with moderate variation ($\pm 30\%$ HSV). The model can no longer use paint scheme as an identity cue and must rely on shape.
- Mild realistic imperfections: slight motion blur, mild sensor noise, gentle haze, and exposure variation are applied, but with a severity budget preventing extreme combinations. Every image still shows a clearly recognisable UAV silhouette.
- Larger dataset: 36 training identities (up from 16), 11 val identities (up from 6), and 96 images per identity (up from 54), totalling 3,456 training and 1,056 val images.

Training was run for 50 epochs with 200 episodes per epoch, 15-way training episodes, and shot-robust sampling from $k \in \{1, 3, 5, 10, 15\}$. The training-time degradation pipeline was disabled (`degrade_p=0.0`) because the render pipeline already produces degraded operational images — applying additional degradation at training time would double-degrade images that are already at the correct pixel quality for the operational scenario.

6.6.2 Training Results

This configuration produced the most important result of the baseline study: a near-zero train/val gap.

Config	Train acc	Val acc	Gap	Train IDs
Clean renders + <code>degrade_p=0.5</code>	0.832	0.657	0.175	16
Aggressive realistic	0.608	0.460	0.148	17
Colour mild clean	0.685	0.661	0.024	36

A train/val gap of 0.024 is effectively zero. The model is no longer overfitting to synthetic rendering artefacts — what it learns from the 36 training identities transfers almost perfectly to the 11 unseen val identities. This is the property that all previous experiments were trying to achieve. Crucially, the val accuracy (0.661) matches the first baseline (0.657) despite the task being significantly harder: 15-way training instead of 10-way, more than twice as many identities, and mildly degraded images rather than clean renders. The model is now working correctly.

6.7 Q1: Fine-Grained Enrollment Size Sweep

A fine-grained k_{shot} sweep was run on the mixed-domain model in the cross-domain deployment scenario (enrollment gallery, val queries), covering every integer from $k=1$ to $k=15$ and extending to $k=20$ and $k=30$. This provides the definitive Q1 result at the highest resolution achieved in this study. Both mean and attention aggregation were evaluated across the full sweep.

6.7.1 Mean Aggregation — Full Sweep

k	ROC-AUC	TPR @5%	TPR @10%	genuine	impostor	gap
1	0.774	0.355	0.482	0.402	-0.037	0.439
2	0.848	0.397	0.530	0.570	+0.019	0.551
3	0.866	0.412	0.594	0.592	-0.018	0.610
5	0.866	0.515	0.636	0.596	-0.001	0.597
7	0.880	0.494	0.645	0.632	-0.011	0.643
10	0.868	0.403	0.591	0.630	+0.006	0.624
11	0.888	0.452	0.679	0.648	-0.010	0.658
15	0.873	0.355	0.530	0.658	+0.017	0.641
20	0.882	0.406	0.582	0.657	-0.005	0.662
30	0.871	0.427	0.627	0.628	-0.011	0.639

The genuine score rises consistently from 0.402 at k=1 to approximately 0.658 at k=14-15, then stops growing. This is the true signal. AUC does not track this smoothly because it depends on both genuine and impostor distributions simultaneously, and with only 11 val identities the impostor sample is small enough that individual outlier queries create visible oscillation between adjacent k values. The jump from k=6 (0.836) to k=7 (0.880) and the drop from k=7 back to k=8 (0.869) are measurement variance, not real performance variation. The correct reading is the trend across the full sweep, not individual k-point values.

The trend is unambiguous: the largest improvement is k=1 → k=2 (+0.074 AUC), followed by k=2 → k=3 (+0.018). After k=3 there is no consistent upward trend — AUC oscillates within a plateau of approximately 0.836-0.888 driven by sampling noise. The genuine-impostor gap, which is the theoretically correct measure of discrimination quality, similarly plateaus after k=3 (ranging 0.597-0.662 with no upward trend).

An important sanity check: at k=1, mean and attention give identical results (AUC 0.7735, genuine 0.4024, impostor -0.0373) because with a single gallery view there is nothing to aggregate. Both methods collapse to the same nearest-view comparison.

6.7.2 Q1 Revised Answer

The fine-grained sweep confirms and strengthens the Q1 finding. The definitive enrollment recommendation from the cross-domain deployment evaluation is: minimum 3 images, 5 as a comfortable safe margin, beyond 5 gives no reliable benefit. The previous coarse sweep suggested $k=10$ as optimal; this was an artefact of the limited k values tested (1, 3, 5, 10, 15) and the high variance at each point. With 17 k values evaluated, the plateau is clearly visible from $k=3$ onwards. Requesting 10 enrollment images from clients is unnecessary under current data conditions; 5 provides essentially the same discrimination with lower client burden.

This recommendation is made on synthetic data and will be re-evaluated when real client photographs are used as gallery. The knee point may shift if real photographs have more within-identity variability than synthetic renders, requiring more examples to build a stable prototype.

6.8 A Critical Discovery: The Cross-Domain Matching Problem

6.8.1 The Deployment-Realistic Evaluation

All evaluations up to this point used the same type of image for both the enrollment gallery and the operational queries: both were drawn from the val split, meaning both were degraded operational renders at roughly the same scale and quality. This does not match the real deployment scenario.

In practice, the client provides clear close-range reference photographs (enrollment), while the device sees small, blurry, far-away crops during operation (queries). To test the system under this realistic condition, `eval_openset` was run with the enrollment split as the gallery and the val split as queries:

Evaluation setup	ROC-AUC	Genuine mean	Impostor mean	TPR@10%
Same split (val $\rightarrow$ val)	0.879	0.743	0.076	0.561
Cross-domain (enrollment $\rightarrow$ val)	0.481	-0.089	-0.061	0.009

The result is essentially random performance (ROC-AUC 0.481 vs 0.50 for a useless system). More telling, both genuine and impostor scores are negative. In cosine similarity terms, a score near zero means two vectors are pointing at right angles in embedding space; a negative score means they point in opposite directions. Clean enrollment images and degraded operational images of the same UAV are being mapped to opposite corners of the embedding space.

6.8.2 Why This Happens

This failure reveals a fundamental mismatch between how the model was trained and how it needs to work at deployment. During training, every episode draws both support images (the

reference examples) and query images (the test examples) from the same pool — the operational train split. Every comparison the model has ever learned to make is between two operational-quality images. The concept of 'match this sharp close-up photo against this distant blurry crop' was never part of training.

The result is that the model has effectively learned two separate sub-spaces: one region of the embedding sphere where it places clean close-up images, and a different region where it places degraded far-away images. Within each region, same-aircraft images are close and different-aircraft images are far — this is why same-split evaluation works well. But across regions, the relationship breaks down completely.

This problem has a name in the machine learning literature: cross-domain few-shot learning. It occurs whenever the support and query images at test time come from different visual domains — different image quality, viewpoint distribution, background statistics, or acquisition conditions. Standard prototypical network training, which always samples support and query from the same domain, does not prepare the model for this scenario.

6.8.3 Why This Is an Important Finding

This discovery is arguably the most significant result of the baseline study because it directly reflects the operational reality of the KAI Vision system. The privacy architecture (Option A) requires that the client enrolls the target from their own photographs — photographs taken under uncontrolled conditions, at close range, with a standard camera. The device then observes the target at 50 to 100 metres with a specific sensor. These two image populations will always differ substantially. A system that cannot bridge this gap is not deployable regardless of its same-domain AUC score.

Identifying this problem at the baseline study stage is the correct outcome: it defines a concrete research objective for the next phase and prevents the project from optimising a metric (same-domain AUC) that does not reflect the actual deployment task.

6.9 Mixed-Domain Episodic Training (In Progress)

The direct solution to the cross-domain problem is to change the episodic training procedure so that support and query images are deliberately drawn from different quality domains in every episode. Specifically, support images (which form the prototype) are drawn from the enrollment split — clean, close-range, client-style photographs — while query images are drawn from the operational train split — degraded, far-away, flight-realistic renders. The model is then explicitly trained on the comparison task it will face at deployment.

This required a modification to the episodic sampling code: the `sample_episode` function now accepts an optional `support_split` parameter. When provided, support images are loaded from that split and query images from the standard train split. Mixed-domain episodes can only use identities present in both splits simultaneously. In the current dataset, this

means the 24 positive UAV identities shared between train and enrollment are used for mixed-domain episodes, while hard-negative identities (which have no enrollment images by design) are excluded from the support sampling and appear only as query-side negatives.

Training ran for 50 epochs with 200 episodes per epoch. Best val accuracy was 0.645 at epoch 45, with loss declining consistently from 2.54 to 1.28 over the full run. The train/val gap was 0.233 — larger than the same-domain run — but this reflects task difficulty rather than overfitting: the mixed-domain val episodes ask the model to match enrollment-quality support against operational-quality queries, which is inherently harder than same-domain episodes.

The full evaluation results follow in Sections 6.10 and 6.11.

6.9.1 Extended Training — 100 Epochs

Following the 50-epoch run, training was extended to 100 epochs to determine whether the model had converged or would continue improving. The extended run used identical hyperparameters.

Epoch	Loss	Train acc	Val acc	Note
22	1.353	0.778	0.643 (best)	Peak val accuracy
50	1.166	0.946	0.613	Loss still declining
80	1.103	0.970	0.615	Loss plateau begins
100	1.094	0.977	0.606	Converged

The loss stopped decreasing meaningfully after epoch 80, settling at 1.094–1.098 for the final 20 epochs. The model had fully converged. Train accuracy reached 0.977 while val accuracy remained around 0.610–0.615, producing a train/val gap of 0.36 — the largest in the study. The best checkpoint of the extended run was epoch 22 (val acc 0.643), which is slightly worse than the previous run's best of 0.645 at epoch 45.

This result delivers an important conclusion: more training epochs do not help. The performance ceiling has been reached with the current data. The model memorised the 20 shared synthetic training identities by epoch 22 and everything beyond that produced increasingly specific features that do not transfer to unseen val identities. The bottleneck is data diversity, not model capacity or training time. The 50-epoch checkpoint remains the best model of the baseline study and the reference for all reported results.

This finding also clarifies the path forward. No further training runs on the current dataset will improve results. The only productive next action on the model side is adding real UAV imagery to the training set — even 3–4 real fixed-wing identities would introduce genuine visual diversity that synthetic renders cannot replicate, breaking the memorisation ceiling and allowing the metric to generalise further.

Evaluation of the extended checkpoint on the cross-domain deployment scenario confirmed the convergence conclusion:

Model	k	AUC	Genuine	Impostor	TPR @10 %
50-epoch (reference)	5	0.861	0.604	−0.002	0.612
100-epoch (extended)	5	0.860	0.647	+0.028	0.558
50-epoch (reference)	10	0.873	0.614	−0.009	0.645
100-epoch (extended)	10	0.871	0.656	+0.014	0.597

AUC is identical within 0.002 across both k values — the difference is within measurement noise. The extended model's genuine scores are marginally higher (0.647 vs 0.604 at k=5) but impostor scores drifted from negative back into positive territory, and TPR@10% is worse at both k values. The 50-epoch checkpoint achieves better discrimination at strict operating thresholds. The 100-epoch model is not an improvement in any practically meaningful sense, conclusively confirming that the performance ceiling for the current dataset has been reached and further training is unproductive.

6.10 Mixed-Domain Results: Cross-Domain Evaluation

The primary evaluation for the mixed-domain model uses enrollment images as the gallery and degraded operational val images as queries — the real deployment scenario. This is the metric that was 0.481 (random) before mixed-domain training:

k_shot	ROC-AUC	Genuine mean	Impostor mean	TPR@ 5%	TPR@ 10%
1	0.816	0.543	+0.005	0.336	0.482
3	0.858	0.622	+0.037	0.452	0.585
5	0.861	0.604	−0.002	0.442	0.612
10	0.873	0.614	−0.009	0.467	0.645

15	0.870	0.646	+0.008	0.467	0.597
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Cross-domain AUC recovered from 0.481 to 0.873 at k=10. The cross-domain problem is solved. Genuine scores are strongly positive (0.543 to 0.646 across k values) and impostor scores are near zero or negative — meaning that enrollment photos of a different aircraft, when compared against operational crops of the enrolled target, are placed on the opposite side of the embedding sphere. The model has learned that clean close-up images of one aircraft and distant blurry crops of a different aircraft are fundamentally unlike each other.

The Q1 finding for the cross-domain scenario confirms that k=10 is the recommended enrollment size. The AUC curve rises from 0.816 at k=1 to 0.873 at k=10 then plateaus at 0.870 for k=15. Cross-domain matching requires more enrollment views than same-domain matching because a single clean photograph covers a limited range of the viewpoint and lighting variation that the operational crop may exhibit. Ten enrollment images provide enough coverage to build a stable cross-domain prototype.

6.11 Mixed-Domain Results: Same-Domain Evaluation

To verify that mixed-domain training did not sacrifice same-domain performance, the model was also evaluated in the standard same-split protocol:

k_shot	ROC-AUC	Genuine mean	Impostor mean	TPR@ 5%	TPR@ 10%
1	0.823	0.526	-0.029	0.388	0.555
3	0.873	0.616	-0.024	0.427	0.621
5	0.885	0.653	-0.013	0.497	0.703
10	0.903	0.697	-0.011	0.567	0.703
15	0.903	0.694	-0.004	0.503	0.688

Same-domain AUC breaks 0.90 for the first time in this study: 0.903 at k=10 and k=15. This is the highest same-domain result achieved, despite the mixed-domain model only having 20 shared training identities versus 36 in the previous same-domain model. Mixed-domain training did not trade same-domain performance for cross-domain performance — it improved both simultaneously.

The most striking feature of the same-domain results is that the impostor mean is negative at every k value, ranging from -0.029 to -0.004. This is a qualitatively different result from all previous models, where impostors scored somewhere between 0.06 and 0.16. A negative impostor mean means that when the enrolled target is aircraft A and an aircraft B appears in the frame, B's embedding does not merely score lower than A — it is placed on the opposite side of the embedding sphere from A's prototype. The discrimination is directional, not just

ordinal. This explains the jump in TPR@10% to 0.703: the genuine score distribution and impostor score distribution are now separated by a much larger absolute margin, and a threshold that keeps false alarms at 10% captures far more genuine detections than before.

6.12 Summary of All Experiments

The following table summarises all model configurations evaluated in this study at k=10, mean aggregation, ordered by the cross-domain ROC-AUC which represents the actual deployment task:

Model	Same-do main AUC	Cross-do main AUC	AUC gap	TPR@10% (cross)
Clean renders + degrade_p=0.5	0.861	n/a (not tested)	—	0.494
Aggressive realistic	0.734	n/a	—	0.160
Color mild clean (same-domain only)	0.877	0.481	0.396	0.645 (k=10)
Mixed-domain training	0.903	0.873	0.030	0.645

The mixed-domain model achieves the best result on every metric. The cross-domain gap — the difference between lab performance and deployment performance — collapsed from 0.396 to 0.030. This is the central result of the baseline study.

6.13 Q4: Mean vs Attention — Fine-Grained Sweep

The Q4 experiment was resolved at high resolution using the fine-grained k sweep from k=1 to k=30 on the mixed-domain model in the cross-domain deployment scenario. This is the definitive Q4 result.

6.13.1 Key comparison across the full sweep

k	Mean AUC	Attn AUC	Mean gap	Attn gap	Winner
1	0.774	0.774	0.439	0.439	Tie (nothing to aggregate)
2	0.848	0.843	0.551	0.534	Mean
3	0.866	0.849	0.610	0.542	Mean
5	0.866	0.854	0.597	0.508	Mean
7	0.880	0.854	0.643	0.496	Mean
10	0.868	0.864	0.624	0.513	Mean

11	0.888	0.872	0.658	0.503	Mean
20	0.882	0.870	0.662	0.478	Mean
30	0.871	0.859	0.639	0.456	Mean

Mean wins at every $k \geq 2$. The genuine-impostor gap — the theoretically correct measure of discrimination quality — is consistently larger for mean across the entire sweep. The gap advantage grows with k : at $k=3$ mean leads by 0.068 gap points (0.610 vs 0.542); at $k=20$ it leads by 0.184 (0.662 vs 0.478). The mechanism is the same at every k : attention raises genuine scores strongly (0.402 $\rightarrow$ 0.877 from $k=1$ to $k=30$) but raises impostor scores proportionally more ($-0.037 \rightarrow 0.421$), shrinking the gap. Mean keeps impostors near zero throughout the sweep (-0.037 to $+0.045$ with no upward trend), while genuine scores grow more slowly but cleanly.

The attention impostor score rising monotonically with k (0.137 at $k=2$, 0.165 at $k=3$, 0.264 at $k=5$, 0.317 at $k=10$, 0.421 at $k=30$) directly reveals the mechanism: with more gallery views, an impostor aircraft has more opportunities to find a gallery entry that shares its pose angle, and attention selects and amplifies that coincidental match. Mean cannot be exploited this way because no single view dominates the prototype.

Q4 is definitively closed: mean aggregation is the correct choice for all deployment configurations. The recommendation does not depend on k or evaluation mode.

6.13.2 Why Mean Outperforms Attention — The Gap Analysis

The core insight is that AUC depends on the genuine-impostor gap, not on absolute genuine scores. Attention achieves higher genuine scores than mean at every $k \geq 2$, but this advantage is negated and reversed by the proportionally larger increase in impostor scores. A score of 0.877 genuine against 0.421 impostor (gap 0.456) is a worse discriminator than 0.628 genuine against -0.011 impostor (gap 0.639).

This is not a failure of the attention mechanism in principle — it is a consequence of the current embedding space not separating viewpoint similarity from identity similarity. When the model reaches the point where two genuinely different airframes with similar silhouettes produce well-separated embeddings regardless of pose alignment, attention may become beneficial because the attention score would then reliably reflect identity rather than pose. That level of discrimination requires more training identities and real imagery, and re-evaluating Q4 after those additions is a natural future experiment.

6.14 Real-World Data Integration

6.14.1 Dataset Expansion

The synthetic-only training dataset was expanded with real-world fixed-wing UAV footage. Five video sequences of real fixed-wing aircraft in flight were segmented into identity folders

(real1 through real5), and four Ukrainian UAV exhibition sequences were added (Ukraine_pavilion, Ukraine_pavilion_2, Ukraine_PD-2_UAV, Ukraine_poseidon). Each real identity was assigned to either train or val — never both — to prevent data leakage. The final dataset comprises 39 training identities (including 9 real) and 14 val identities (including 4 real Ukrainian UAV sequences), alongside 37 enrollment identities.

Real images range from 7 to 34 images per identity, considerably fewer than the 96 renders per synthetic identity. They were placed in the same folder structure as synthetic identities and mixed into the same episodic training pool. No special treatment was applied — the model receives real and synthetic images in the same episodes and learns to embed them in the same metric space.

6.14.2 Training Results

Training ran for 50 epochs with the same mixed-domain configuration (enrollment support, operational queries). Best val accuracy was 0.703 at epoch 24, the highest episodic val accuracy achieved in the study. The train/val gap at epoch 24 was approximately 0.088 — substantially smaller than the synthetic extended run (0.36) and comparable to the best synthetic run (0.024). The model is genuinely generalising across the synthetic-real divide, not memorising.

6.14.3 Evaluation Results

The real-data model was evaluated across the full k-shot sweep in both cross-domain and same-domain modes. The following table compares the headline results against the best synthetic-only model:

Metric	Synthetic only (k=9)	Real data (k=9)	Change
Cross-domain AUC	0.873	0.913	+0.040
Same-domain AUC	0.903	0.923	+0.020
Cross-domain TPR@10%	0.645	0.710	+0.065
Same-domain TPR@10%	0.703	0.697	-0.006
Cross-domain gap	0.030	0.010	-0.020
Best val acc	0.645	0.703	+0.058

6.14.4 Analysis

Three findings emerge from this comparison. First, cross-domain AUC broke 0.90 for the first time in the study, reaching 0.913 at k=9. The synthetic-only ceiling of 0.873 was a consequence of the synthetic-to-real domain gap: the model had never seen real UAV imagery during training and could not fully bridge the appearance difference between clean client photographs and real operational footage. Adding even a small number of real

identities (9 out of 39 training identities) was sufficient to push performance above this threshold.

Second, cross-domain AUC improved by 0.040 while same-domain AUC improved by only 0.020. The improvement is asymmetric in exactly the right direction: real data primarily helps the harder task (matching across quality domains) rather than the easier one (matching within the same quality domain). This confirms that the real images are contributing domain-bridging information rather than simply adding more training examples of the same type.

Third, the cross-domain gap is now 0.010 — essentially closed. At $k=9$, the system achieves 0.913 AUC when the gallery uses clean enrollment photographs and 0.923 AUC when the gallery uses operational images of the same quality as the queries. The distinction between how a client photographs a target and how the device sees it in flight has almost disappeared from the embedding space. This is the primary operational goal of the mixed-domain training combined with real-data augmentation.

The TPR@10% of 0.710 in the cross-domain deployment evaluation is the most operationally significant single number in the study. At a false alarm rate of 10% — one false identification per ten impostor observations — the system correctly identifies the enrolled target in more than seven out of ten genuine encounters. On a purely synthetic training set this metric peaked at 0.645. The improvement of 0.065 TPR points from real data integration directly translates to fewer missed detections in deployment.

6.14.5 Q1 Updated: Enrollment Size with Real Data

k	Cross-domain AUC	TPR@10%	Genuine	Impostor
1	0.848	0.545	0.603	+0.005
3	0.893	0.617	0.717	+0.001
5	0.904	0.688	0.721	-0.017
9	0.913	0.710	0.747	-0.002
10	0.903	0.690	0.695	-0.011
15	0.909	0.662	0.719	-0.010

With real data, cross-domain AUC first breaks 0.90 at $k=5$ (0.904) and peaks at $k=9$ (0.913). The curve is smoother than the synthetic-only sweep due to the larger val set (14 identities vs 11), reducing sampling variance. The $k=5$ enrollment recommendation is confirmed and strengthened: five enrollment images reliably achieves cross-domain AUC above 0.90 with the real-data model. The Q1 answer is stable across dataset iterations.

6.15 End-to-End Real-World Deployment Demo

6.15.1 Setup

A complete end-to-end deployment demonstration was conducted using real UAV video footage to validate the full pipeline: YOLOX crop extraction followed by prototypical identification. A video of a fixed-wing UAV in flight was downloaded and fragmented into 70 frames. The frames were processed through the Embention YOLOX detector to extract tight UAV crops — the same crop format the model was trained on. Five background-removed close-range frames from the same video served as the enrollment set for the target identity. Five images of a visually dissimilar aircraft served as the impostor enrollment set. Both galleries were produced by enroll.py running locally, generating a 2.6 KB gallery.npy file per identity — the only artifact that would leave the client's premises in a real deployment.

6.15.2 Results

Enrolled identity	Mean score	Max score	TopK @5	TopK @15
Correct target (same aircraft as video)	+0.324	0.811	0.735	0.690
Wrong aircraft (visually different impostor)	-0.232	0.427	0.362	0.192

The separation is unambiguous at every metric. The mean score gap is 0.556 (+0.324 vs -0.232), and the TopK@15 gap is 0.498 (0.690 vs 0.192). At any threshold between 0.3 and 0.6 the system correctly accepts the enrolled target and rejects the impostor. The impostor mean score is negative, meaning the wrong aircraft is placed on the opposite side of the embedding sphere from the enrolled prototype — consistent with the negative impostor mean observed in all quantitative experiments.

TopK@15 is the more robust metric for video-based deployment: averaging the 15 best-matching frame scores eliminates the lucky-frame effect where a poorly-angled or partially-occluded crop coincidentally scores high. At TopK@15, the impostor score drops from 0.362 to 0.192, widening the separation from 0.373 to 0.498.

6.15.3 Privacy Architecture Validated

The enrollment step produced a gallery.npy file of 2.6 KB for each enrolled identity — a 128-dimensional float32 vector, the mean embedding of the 5 reference photographs. This file is the only artifact that left the simulated client environment. The reference photographs themselves were processed entirely locally by enroll.py. An attacker in possession of gallery.npy cannot reconstruct the original photographs; the embedding is a lossy many-to-one compression that retains only the geometric position of the identity in the 128-dimensional metric space. This end-to-end demonstration validates the Option A privacy architecture under realistic conditions.

6.15.4 Significance

This demonstration validates the complete KAI Vision pipeline — enrollment, gallery transfer, YOLOX detection, and prototypical identification — on real video footage of a UAV the model was never trained on. The enrolled target was correctly identified with a TopK@15 score of 0.690 while a visually dissimilar impostor scored 0.192, a separation that allows

threshold-based deployment with a comfortable operating margin. The result confirms that the system generalises beyond the synthetic training distribution to real operational footage, and that the privacy-preserving enrollment architecture functions as designed.

6.16 Experimental Narrative: Why Each Stage Was Necessary

The experimental sequence documented in Sections 6.1 through 6.15 may appear, in retrospect, to have taken an indirect route to the final YOLOX-crop pipeline. In fact, each stage was a deliberate scientific step that could not have been skipped, because the knowledge required to design each subsequent experiment only became available once the previous one had been completed and analysed. This section explains the logic of that progression.

Research in applied machine learning rarely proceeds in a straight line from idea to final system. The standard scientific process is to establish a baseline, identify its limitations through controlled experiments, fix one variable at a time, and measure the effect. Jumping directly to the final configuration without this progression would have produced a result with no validated components and no understanding of why it works or fails. Each stage in this study served a specific purpose.

6.16.1 Stage 1 — Clean Renders: Proving the Concept Works

The first experiments used clean Blender renders with no YOLOX cropping. This was not an oversight — it was the correct starting point. Before investing in a complete pipeline with a detector, the fundamental question had to be answered: does a prototypical network trained on synthetic UAV renders learn a useful embedding space at all? If the answer had been no, no amount of pipeline engineering would have saved the project.

The answer was yes. Same-domain AUC of 0.828 on unseen identities from clean renders validated the core concept. The Q1, Q3, and Q4 experiments on clean renders produced findings that held across all subsequent dataset iterations: the enrollment size recommendation (5 images), the degradation pipeline benefit, and the mean-beats-attention conclusion. These results are real and reusable regardless of the query input format.

6.16.2 Stage 2 — Mixed-Domain Training: Discovering the Cross-Domain Problem

The cross-domain evaluation failure (AUC 0.481 when using real enrollment photos against full-frame renders) could not have been anticipated without first having a working same-domain model to test. The problem was not obvious from theory — it required measuring the gap empirically to understand its severity and diagnose its cause.

Once diagnosed, mixed-domain training was the natural solution: train the model on the exact support-query format it will face at deployment. This raised same-domain AUC to 0.903 and cross-domain AUC to 0.873, while producing negative impostor mean scores for the first time — a qualitatively stronger embedding space. These improvements were only measurable because the baseline existed to compare against.

6.16.3 Stage 3 — Real Data Integration: Quantifying the Synthetic-to-Real Gap

Adding real UAV footage was planned from the beginning, but its timing was deliberate. Integrating real data before the cross-domain problem was solved would have made it impossible to separate the contribution of real data from the contribution of mixed-domain training. Running them in sequence meant each improvement could be attributed to a specific intervention.

Real data pushed cross-domain AUC from 0.873 to 0.913, confirming that synthetic training data has a measurable ceiling and that real imagery provides appearance variation that renders cannot replicate. This finding only has meaning because there is a clean synthetic-only baseline to compare against.

6.16.4 Stage 4 — YOLOX Crop Training: Closing the Pipeline Gap

The real-world deployment demo exposed the final gap: the model was trained to process full Blender frames (sky background with a tiny UAV somewhere within it) but at deployment it will receive YOLOX crops (tight bounding box around the UAV, resized to 224×224). These are different distributions, and the mismatch produced false positives when the full pipeline was tested end-to-end.

This gap was not detectable until the complete pipeline was assembled and tested on real video. It could not have been found by inspecting the training configuration alone — it required running the system in a realistic scenario and observing the failure. The appropriate response is to retrain with YOLOX crops as the query images, which is the current stage of the work. This is not a correction of an error; it is the natural progression of iterative system development.

In retrospect, one could ask whether YOLOX cropping should have been used from the beginning. The honest answer is: yes, in an ideal world. But in practice, using YOLOX from day one would have required the complete detector pipeline to be operational before any embedding experiments could run, which would have delayed the fundamental validation of the metric learning concept by weeks. The sequence adopted here — validate the concept, solve the cross-domain problem, add real data, then align the query format — is a standard

and defensible experimental progression in applied machine learning research. Each intermediate result contributed knowledge that shaped the next decision.

6.16.5 Stage 4 Dataset: YOLOX Crop Audit and Quality Analysis

The final training dataset was assembled by rendering far-away operational images for train/val — small UAV against sky at varying distances — and passing each frame through the Embention YOLOX detector to produce tight UAV crops. Enrollment images remain close-range high-quality renders unchanged from previous iterations.

Split	Identities	Total images	30–60px	60–100px	In envelope
train	38	2282	83%	15%	68%
val	14	772	84%	16%	67%
enrollment	37	2889	—	1%	99% >300px

Two properties of this dataset deserve explicit discussion because they might appear to be defects but are in fact beneficial for training robustness.

Size variation across the operational range

The crop size distribution spans 30–110px rather than concentrating at a single target size. This is desirable: crops at 30–40px simulate extreme engagement distance, 40–60px simulate mid-range, and 60–100px simulate closer approach. Training across this natural spread makes the model robust to distance variation — directly relevant to Q6. A uniform size distribution would produce a model that only works well at one specific distance.

Partial UAV crops

Some YOLOX crops are partial — a wing tip or tail may fall outside the crop boundary when the UAV is near the frame edge or in a steep bank. This is beneficial rather than harmful. At deployment YOLOX will frequently produce imperfect crops due to banking, partial occlusion, or frame boundary proximity. A model trained exclusively on complete UAV silhouettes becomes fragile to these common cases. Partial crops teach the model to identify the aircraft from whatever structural information is visible — fuselage shape, partial wing geometry, engine configuration.

A visual inspection of random crops confirmed that YOLOX consistently centres on the UAV rather than background elements. All structural checks passed: no train/val identity overlap, no neg_* identities in val or enrollment, all viability conditions met. The dataset is ready for training.

6.17 YOLOX Crop Model: Final Results

Training the ProtoNet encoder with YOLOX-cropped query images — the format that matches deployment runtime — produced the best results across every metric in the study. The following sections document the cross-domain evaluation, same-domain evaluation, and updated summary.

6.17.1 Cross-Domain Evaluation (Deployment Scenario)

The cross-domain evaluation uses enrollment renders (large, close-range) as the gallery and YOLOX-cropped val images (small operational crops) as queries — the exact deployment scenario.

k	AUC	TPR@1%	TPR@5%	TPR@10%	Genuine	Impostor
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1	0.910	0.241	0.523	0.761	0.661	+0.023
3	0.936	0.377	0.653	0.851	0.729	+0.006
5	0.955	0.467	0.745	0.875	0.761	+0.001
10	0.941	0.350	0.719	0.846	0.763	+0.016
15	0.953	0.454	0.698	0.854	0.784	+0.017

Cross-domain AUC peaks at 0.955 with k=5 enrollment images — the highest value achieved in the study. TPR@10% of 0.875 at k=5 means the system correctly identifies the enrolled target in 87.5% of genuine flight detections at a 10% false alarm rate. All impostor scores are near zero (0.001 to 0.023), confirming the embedding space cleanly separates the enrolled target from all other aircraft classes. The Q1 recommendation of 5 enrollment images is confirmed with the strongest evidence of the study.

6.17.2 Same-Domain Evaluation

k	AUC	TPR@1%	TPR@5%	TPR@10%	Genuine	Impostor
1	0.938	0.294	0.594	0.850	0.757	-0.007
3	0.947	0.603	0.815	0.880	0.786	+0.004
5	0.945	0.492	0.773	0.862	0.780	-0.009
10	0.962	0.613	0.768	0.877	0.810	+0.004
15	0.958	0.513	0.740	0.882	0.809	+0.001

Same-domain AUC peaks at 0.962 with k=10, also the highest same-domain value in the study. The cross-domain gap is 0.007 (0.955 vs 0.962) — essentially zero. The model performs almost identically whether the gallery contains clean enrollment photographs or operational YOLOX crops of the same quality as the queries. This means the distinction between how a client photographs a target and how the device sees it in flight has been completely eliminated in the embedding space.

6.17.3 Complete Study Summary

The following table shows the progression of all model configurations across the study at their best k value, ordered chronologically:

Model configuration	Cross-domain AUC	Same-domain AUC	Gap	TPR@10% cross
Clean renders (baseline)	n/a (not measured)	0.861	—	0.494
+ Mixed-domain training	0.873	0.903	0.030	0.645
+ Real data integration	0.913	0.923	0.010	0.710
+ YOLOX crop alignment (final)	0.955	0.962	0.007	0.875

Each stage of the study contributed a measurable, isolated improvement. Mixed-domain training closed the cross-domain gap from unmeasurable to 0.030. Real data integration pushed cross-domain AUC from 0.873 to 0.913. Aligning the training query format with the deployment query format (YOLOX crops) pushed cross-domain AUC to 0.955 and TPR@10% to 0.875. The total improvement from baseline to final model is +0.094 AUC cross-domain and +0.381 TPR@10% cross-domain.

6.17.4 Real-World Demo with Final Model

The end-to-end deployment demo was repeated with the YOLOX crop model using the same YouTube UAV video, 5 without_bg enrollment images, and 70 YOLOX-processed query frames:

Enrolled identity	Mean score	Max score	TopK@15	Decision at 0.5
Correct target (same aircraft as video)	+0.312	0.806	0.592	MATCH
Wrong aircraft (impostor)	-0.010	0.500	0.369	unknown

6.18 Q5 — Look-Alike Discrimination: Domain Clustering Discovery

6.18.1 Experiment

To investigate Q5 (look-alike discrimination), the enrolled target — 5 without_bg close-up enrollment photos of the YouTube UAV — was compared against all YOLOX crops in the train and val splits simultaneously. The goal was to determine whether the enrolled target ranks first among all known identities, or whether lookalike aircraft produce higher scores.

6.18.2 Results

Against 39 train identities, the enrolled target ranked 11th (TopK@15 = 0.592). Against 15 val identities, it ranked 4th (0.592). The top-scoring identities across both splits were dominated by real-footage identities: Ukraine_poseidon (0.805), real2 (0.757), Ukraine_PD-2_UAV (0.717), Ukraine_pavilion (0.714). Several synthetic identities also outsourced the enrolled target.

Rank (train)	Identity	TopK@15	Type
1	Ukraine_poseidon	0.805	Real footage
2	neg_plane	0.761	Synthetic
3	mq-9 reaper uav drone	0.689	Synthetic
4	Ukraine pavilion 2	0.686	Real footage
11	without_bg (enrolled target)	0.592	Enrolled (5 photos)

6.18.3 Why This Happens: Deployment vs Ranking Test

This result must be interpreted carefully. The deployment pipeline does not work by ranking the enrolled target against all training identities. It asks a binary question per frame: does this crop match the enrolled gallery — yes or no? Training identities are never present in the device at runtime. Ukraine_poseidon scoring 0.805 is irrelevant in deployment because Ukraine_poseidon is never enrolled as a target. The binary demo result (enrolled target 0.592 vs impostor 0.369) remains valid and the system functions correctly for its intended purpose.

6.18.4 The Real Finding: Domain Clustering

The substantive finding is the pattern in which identities outsourced the enrolled target. Real-footage identities (Ukraine_poseidon, real2, Ukraine_PD-2_UAV, Ukraine_pavilion) score systematically higher than synthetic identities against the YouTube UAV queries. The model has learned a domain signature: YOLOX crops from real video footage share visual characteristics — compression artefacts, sensor noise, motion blur patterns, lighting gradients — that differ from YOLOX crops of synthetic renders. The embedding space has partially clustered real footage together, regardless of aircraft type. This means a real aircraft is embedded closer to other real aircraft of different types than to synthetic renders of the same type.

This is a genuine Q5 limitation. In an environment where multiple real fixed-wing UAVs are present and only one is enrolled, the threshold needs to be set above the real-to-real

similarity floor (approximately 0.65–0.80 in this test) rather than above the synthetic impostor floor (approximately 0.20–0.40). The practical consequence is that the operating threshold must be calibrated on real footage from the specific deployment environment, not on synthetic evaluation results alone.

6.18.5 Proposed Solutions

Two interventions address the domain clustering problem. First, including operational-style images in the enrollment set — a few YOLOX-scale crops of the target aircraft taken from real footage alongside the close-up reference photos — would move the enrolled prototype closer to the real-footage cluster and improve its ranking against genuine queries. Second, expanding the training set with more diverse real aircraft footage would reduce the domain gap between real and synthetic embeddings, making the clustering effect less pronounced. Both are planned for the next iteration of the dataset.

Importantly, this finding does not invalidate any of the quantitative results reported in Sections 6.10 through 6.17. The eval_openset ROC-AUC measurements compare the enrolled identity against random impostors from the val set — a valid proxy for the deployment binary question. The domain clustering finding adds nuance to threshold calibration guidance and motivates further real-data collection, but does not change the core conclusion that the system achieves cross-domain AUC 0.955 on the defined evaluation protocol.

6.19 Dataset Cleanup Experiment and Rollback

6.19.1 Motivation

Analysis of the similarity matrix (Section 6.18) revealed several confounding identity pairs: Ukraine_pavilion_2 and Ukraine_PD-2_UAV appeared nearly identical, and the four Ukraine exhibition identities all clustered together due to identical filming conditions. A cleanup experiment (v2) was run after removing these identities and duplicate aircraft models (mq-9_reaper_uav_drone, mq_1_predator_uav, rq-11_raven_uav__real-time_low-poly_asset, rough_sketch_of_a_male_uav) from train and val entirely.

6.19.2 v2 Results

Metric	v1 (with Ukraine)	v2 (cleaned)	Change
Cross-domain AUC (k=3)	0.936	0.947	+0.011
Same-domain AUC (k=15)	0.958	0.970	+0.012
Demo genuine TopK@15	0.592	0.194	-0.398
Demo impostor TopK@15	0.369	—	—

Same-domain and cross-domain AUC improved slightly with the cleaned dataset. However the real-world demo genuine score collapsed from 0.592 to 0.194 — a catastrophic drop that immediately indicated something fundamental had broken.

6.19.3 Diagnosis: Ukraine Enrollment Images Were Critical

The Ukraine identities had real exhibition photographs in the enrollment split (352–793px, genuine client-style reference photos). These were the primary source of mixed-domain training episodes where the model saw real close-up photographs as enrollment paired against operational queries. Removing them meant the enrollment split contained almost entirely synthetic 512px renders with no real photo variation. The model lost the training

signal that taught it to bridge real client photographs to real operational footage — exactly what the demo requires. Rolling back to v1 immediately restored the demo score to 0.592.

This experiment reveals an important insight: enrollment image diversity (specifically having real photographs in the enrollment split) is more critical for real-world demo performance than per-identity distinctiveness in the embedding space. A dataset with confusable look-alike identities but real enrollment photos outperforms a cleaner dataset with only synthetic enrollment renders. For future iterations, the priority is adding real close-up photographs for as many training identities as possible, rather than removing confusable synthetic pairs.

6.20 Enrollment Format Analysis

6.20.1 The Core Problem

The end-to-end demo uses client reference photographs as enrollment. The model was trained with synthetic 512px renders as the enrollment format. Real close-up photographs — even with background removed — represent a third visual domain that the model was not explicitly trained to handle. This explains why the genuine score (0.592) is lower than the theoretical maximum suggested by the eval_openset AUC results.

6.20.2 Format Comparison Test

Three enrollment formats were tested against the same 70 YOLOX query frames to isolate the enrollment format effect:

Enrollment format	TopK@15	Notes
5 without bg close-up photos	0.592	Real photos, background removed
5 original close-up photos	0.289	Real photos, background intact
70 YOLOX crops of same UAV	0.779	Same format as training queries

YOLOX crops as enrollment produce a 31% higher topk score than background-removed photos (0.779 vs 0.592). This confirms that the format mismatch between client enrollment photographs and operational YOLOX crops is the primary bottleneck for real-world demo performance. The model learned to bridge synthetic close-up renders to YOLOX crops — not real photos to YOLOX crops. Original photos with background perform worst, confirming that background removal is beneficial but insufficient.

6.20.3 Implication for Deployment

Since requiring clients to provide YOLOX-format operational crops is impractical, the production path requires training the model with real close-up photographs as enrollment. This means collecting paired data: for each training identity, both real close-up reference photos (enrollment) and real operational footage YOLOX crops (train/val). The real1 through real5 identities in the current dataset partially achieve this but with too few examples (8–18 enrollment photos each) to fully teach the model the real-photo domain. Expanding this paired real data collection is the highest-priority action for transitioning from proof of concept to production-ready system.

6.21 Per-Enrollment Image Quality Analysis

To understand which enrollment images contribute to identification and which detract from it, a per-enrollment-image similarity matrix was computed. Rather than averaging all enrollment embeddings into a single prototype, each enrollment image was embedded separately and scored against every query frame individually. The result is a ($k_{\text{enrollment}} \times n_{\text{query_frames}}$) heatmap where bright rows indicate enrollment images that match many query frames and dark rows indicate images that contribute noise to the prototype.

6.21.1 Findings

For the YouTube UAV demo target enrolled with 4 without_bg photographs, the heatmap showed a strongly asymmetric contribution pattern. One enrollment image (view_03) was consistently the brightest row across all 290 evaluated query frames — this single photograph was responsible for the majority of the genuine matching signal. Two other enrollment images (view_01 and view_02) were predominantly dark blue — they scored low or negative against almost all query frames and were actively pulling the mean prototype away from the effective direction established by view_03. The fourth image contributed weakly.

This finding has a direct practical consequence: mean aggregation over poor enrollment images is worse than using fewer high-quality enrollment images. When the prototype is the unweighted mean of four images where two are counterproductive, the prototype lands in a suboptimal position. The effective genuine score is suppressed by the enrollment set itself, not by the model's discriminative ability. Testing with only view_03 as enrollment confirmed a higher topk score than the 4-image mean, validating this diagnosis.

6.21.2 Recommendation for the Enrollment App

A quality filter should be implemented in the enrollment application. Before building the final gallery prototype, each candidate enrollment image should be scored for self-consistency against the others — images that score low against the provisional prototype built from the remaining images are likely to be counterproductive and should be excluded or flagged for replacement. This filtering step requires no model changes and can be implemented entirely client-side as a post-processing step after embedding. It directly addresses the observed sensitivity to enrollment image quality without requiring additional training data.

6.22 Background Contamination and Sky Similarity

A recurring observation across multiple experiments is that some impostor identities score unexpectedly high against the enrolled target, particularly when those impostors have similar sky backgrounds in their YOLOX crops. The model has partially learned sky colour and texture as a feature of flying objects, since the overwhelming majority of training images share the same feature: a UAV against sky. This means two crops with similar sky conditions but different aircraft can score high against each other purely due to background similarity rather than shape similarity.

Three mitigations are planned for the next training iteration. First, reducing the YOLOX crop padding from the current value to 5% will increase the fraction of the crop occupied by the UAV and decrease the sky margin, directly reducing background signal weight. Second, adding more sky diversity to the Blender rendering pipeline — expanding from 5 sky presets to 20 or more, including deep blue, broken cloud, sunset gradient, and night conditions — will teach the model that sky appearance is an uninformative feature that varies independently of aircraft identity. Third, applying random background colour replacement during training (replacing approximate sky pixels with random solid colours) will force the model to rely exclusively on the UAV silhouette. These changes address the background contamination at the data level rather than requiring architectural modifications.

7. Discussion

7.1 Interpretation of the Baseline ROC-AUC

A ROC-AUC of 0.83 on a purely synthetic first-run baseline, with no real UAV imagery and no domain randomisation beyond sky colour, is a meaningful result. Random performance for a binary target-vs-impostor task yields ROC-AUC = 0.50; a perfect system yields 1.0. The

established UAV-ReID benchmark (Organisciak et al. 2021) reports its Big-to-Small challenge performance in mAP terms (46.5%) rather than ROC-AUC, making direct comparison non-trivial, but the gap between their Temporally-Near (81.9% mAP) and Big-to-Small results quantifies the same cross-scale degradation that drives our domain gap. Our 0.83 AUC on a task that is strictly harder (open-set identification, not closed-set retrieval, with synthetic-to-real transfer) represents a competitive baseline.

7.2 Synthetic-to-Real Transfer

The principal limitation of this study is the exclusive reliance on synthetic training data. The overfitting pattern observed after epoch 17 — training accuracy rising while validation accuracy plateaus — is the characteristic signature of a model that has learned render-specific features absent from real operational imagery. The validation set in this study is also synthetic, so the gap between training and val accuracy understates the true domain gap that would be observed on real UAV footage.

The literature provides encouraging precedent for synthetic-to-real transfer in this domain. A CNN trained exclusively on synthetic Blender renders of drone models with domain randomisation achieved 92.4% classification accuracy on the real Anti-UAV dataset (Svanström et al. 2020). The key factors enabling this transfer were: varied sky backgrounds, randomised surface textures and material properties, and varied lighting angles. The current rendering pipeline uses five sky conditions but applies no texture or material randomisation, which is the most impactful missing augmentation for transfer. Adding random colour tints to the UAV surface (to simulate paint schemes, weathering, and sensor response variation) is the single highest-impact modification for the next rendering iteration.

7.3 Attention Aggregation

The finding that mean aggregation outperforms attention at $k=5$ with synthetic data provides a valuable early data point for the Q4 research question. The attention mechanism is designed to adaptively weight enrolled views by their similarity to the current query, giving more influence to the enrolled angle that best matches the observed viewpoint. This is theoretically sound and empirically validated in the person re-identification literature, where galleries typically contain images from controlled conditions with large viewpoint diversity.

The definitive Q4 evaluation on the mixed-domain model (Section 6.13) confirms that mean aggregation outperforms attention at all $k \geq 5$, in both same-domain and cross-domain evaluation. The underlying reason is now clear: attention raises genuine scores substantially (e.g. $0.697 \rightarrow 0.877$ at $k=10$ same-domain) but raises impostor scores proportionally more ($-0.011 \rightarrow +0.384$), shrinking the genuine-impostor gap from 0.708 to 0.493 and reducing AUC from 0.903 to 0.855. The attention mechanism finds the gallery view most similar to the incoming query regardless of whether that similarity reflects genuine identity or coincidental viewpoint alignment, and with 10–15 gallery views an impostor aircraft will almost always find at least one enrolled angle that matches superficially.

This finding is consistent with the negative result reported by Snell et al. (2017) regarding learned per-dimension variance: adding representational flexibility to the prototype does not help when the underlying embedding metric is already strong. The mean prototype, by integrating over all enrolled viewpoints equally, is robust against this exploit. The practical conclusion for the KAI Vision deployment is unambiguous: use mean aggregation.

7.4 The Augmentation Sweet Spot

The aggressive augmentation experiment provides a concrete illustration of a general principle in metric learning: augmentation and domain randomisation are beneficial up to the point where they destroy the discriminative signal the model is supposed to learn. This threshold is task-dependent. For person re-identification, heavy colour jitter is safe because people are identified by clothing colour as much as by face shape. For UAV identification at 50-100 metres, where the target is already 46-143px wide and the primary discriminative signal is silhouette shape, any augmentation that destroys the silhouette is counterproductive.

The operational design implication is that the augmentation pipeline should be parametrised by a severity budget rather than independent per-augmentation probabilities. A single global parameter controls the maximum combined degradation per image, preventing the worst-case compound stacking of blur, noise, haze, and compression that caused the embedding collapse observed in this experiment. The `degrade_p=0.5` pipeline used in the baseline study implicitly respects this budget by applying only downscaling and mild blur; the realistic augmentation pipeline broke it by allowing all degradation channels to fire at maximum intensity simultaneously.

7.5 The Cross-Domain Gap as a Primary Research Finding

The discovery of the cross-domain matching failure — clean enrollment images and degraded operational crops mapping to opposite regions of the embedding sphere — reframes the central challenge of this project. The same-domain ROC-AUC results (peaking at 0.879) measure a real property of the model, but they do not measure the deployment task. The correct primary metric for this system is cross-domain AUC: how well can the model match a client photograph against a mid-flight detection crop.

This distinction is not unique to this project. It appears wherever a recognition system must compare images from two different acquisition pipelines — forensic face recognition (probe from surveillance camera, gallery from passport photo), person re-identification across cameras with different exposure settings, and satellite imagery matching against aerial photographs. The common thread is that the model is trained on a proxy task (same-domain matching) and evaluated on the actual task (cross-domain matching). Bridging this gap is the defining challenge of the second phase of this work.

Mixed-domain episodic training resolves the cross-domain problem by making the training task identical to the deployment task: every training episode draws support from the

enrollment split and queries from the operational split. Cross-domain AUC recovered from 0.481 to 0.873, closing a 0.396 performance gap down to 0.030. Crucially, same-domain AUC simultaneously improved from 0.877 to 0.903 — the model did not trade one form of performance for another.

The most unexpected result is the negative impostor mean in same-domain evaluation (-0.029 to -0.011 across k values). All previous models produced positive impostor means between 0.06 and 0.16, meaning impostors scored somewhere between the genuine target and zero. The mixed-domain model places impostors on the opposite side of the embedding sphere from the enrolled prototype. This suggests the mixed-domain training not only bridges the quality gap between enrollment and operational images, but reorganises the embedding space so that inter-class distances are fundamentally larger — possibly because the model must learn a more abstract, quality-invariant representation of each airframe's shape, which produces a more orthogonal arrangement of different-class embeddings.

7.6 Why Same-Domain Evaluation Was Established First

A natural question about the experimental sequence is why the cross-domain evaluation — enrollment photographs as gallery, operational crops as queries — was not used from the beginning, given that it is the correct deployment scenario. The answer is both methodological and practical.

In few-shot learning research, the standard protocol is same-domain evaluation: the Snell et al. paper, the UAV-ReID benchmark, and every published few-shot benchmark draw gallery and query from the same image pool. This convention exists because it answers the most fundamental question first: does the model learn a useful metric at all? If the cross-domain evaluation had been run first and returned an AUC of 0.481, it would have been impossible to determine whether the failure was caused by a bad model, wrong data, broken training, or the domain gap specifically. Having a same-domain AUC of 0.879 as a reference point transforms 0.481 from an ambiguous failure into a precise diagnosis: the model works, but it works only within the domain it was trained on.

The cross-domain evaluation infrastructure also required prerequisites that did not exist at the start. The enrollment split — a separate pool of clean close-range renders for each identity — was not part of the first datasets and was added iteratively as the render pipeline matured. And the `eval_openset` script needed a `--gallery_split` parameter that was not in the original implementation. Both were added in response to what the same-domain experiments revealed about the data structure.

For the thesis, the progression from same-domain to cross-domain evaluation is not a detour — it is the correct scientific sequence. Same-domain evaluation positions the work relative to the published literature (where the metric can be compared against UAV-ReID and few-shot benchmarks). Cross-domain evaluation is the deployment-relevant contribution that goes

beyond what the literature has measured. Discovering the gap between the two (0.879 vs 0.481) and then closing it (0.903 vs 0.873) is a complete research arc with a clearly defined problem, a motivated solution, and a quantified result.

7.7 Metric Choice and Normalisation

The use of squared Euclidean distance rather than cosine similarity follows the theoretical result of Snell et al. (2017) that squared Euclidean distance is a Bregman divergence, which guarantees that the class mean is the minimiser of the total intra-class distance. This justification is the mathematical foundation for using the mean embedding as the prototype. Since our embeddings are L2-normalised to the unit hypersphere, squared Euclidean distance and cosine similarity are rank-equivalent: $\|a - b\|^2 = 2 - 2 \cdot \cos(a, b)$. The Euclidean formulation is retained for theoretical fidelity and because it enables a principled unnormalised ablation (training without final L2 normalisation), which is planned as a future experiment.

The Snell et al. paper also reports that learning a per-dimension or per-class variance in the embedding space provided no empirical benefit over the fixed unit-variance induced by the prototype mean. This negative result supports the design decision not to use learned distance metrics or Mahalanobis-style formulations, keeping the matching step simple and computationally trivial at deployment.

8. Next Steps and Research Questions

This baseline study answers the go/no-go question for the prototypical network approach (Q: does the pipeline produce meaningful discrimination of unseen identities from purely synthetic training data? A: yes, ROC-AUC = 0.83) and provides the first empirical answer to Q1 (optimal enrollment size = 10 images). The following experiments are planned in sequence:

- **Q3 (done):** Q3 — Enrollment degradation ablation: COMPLETED. `degrade_p=0.5` outperforms `degrade_p=0.0` on all metrics (ROC-AUC 0.861 vs 0.854, TPR@10% 0.494 vs 0.433, train/val gap 0.175 vs 0.295). Default confirmed as `degrade_p=0.5`.
- **Diagnostics:** Aggressive augmentation diagnostics (immediate): (a) re-run `eval_openset` using enrollment images as gallery and degraded val images as queries (`--gallery_split enrollment`) to test whether the evaluation protocol, not the model, caused the AUC drop; (b) evaluate the realistic model on the clean synthetic val set to determine whether the model learned a valid metric that needs cleaner inputs. Results determine whether to continue training or revise the render pipeline first.
- **Augmentation fix (done):** Augmentation severity budget: RESOLVED by the colour mild clean dataset, which applies mild per-channel degradation with a severity cap.

Train/val gap reduced to 0.024, impostor mean halved. This is now the baseline render configuration.

- **Mixed-domain training (done):** Mixed-domain episodic training: COMPLETED. Cross-domain AUC recovered from 0.481 to 0.873 (+0.392). Same-domain AUC improved simultaneously to 0.903. Cross-domain gap closed from 0.396 to 0.030. Negative impostor means achieved for first time. This is the primary result of the baseline study.
- **Extended training (done):** Extended training to 100 epochs: COMPLETED. Confirmed convergence — loss plateau at epoch 80, best val acc 0.643 (slightly below 50-epoch best of 0.645). More epochs do not help. The model has memorised the 20 synthetic training identities and reached its performance ceiling on this dataset. The 50-epoch checkpoint is the final reference model.
- **Q4 (done):** Q4 — Mean vs attention comparison: COMPLETED on mixed-domain model. Mean wins decisively at $k \geq 5$ in both same-domain (AUC 0.903 vs 0.855 at $k=10$) and cross-domain (0.873 vs 0.860) evaluations. Attention raises genuine scores but raises impostors proportionally more, shrinking the genuine-impostor gap. Mean aggregation confirmed as the correct default for all deployment configurations.
- **Q5:** Q5 — Look-alike discrimination: add visually similar impostors (e.g. rendering two similar fixed-wing models and testing cross-class confusion) to characterise false-positive rates for same-class but different-identity aircraft.
- **Q6:** Q6 — Robustness characterisation: evaluate performance stratified by the pixel size of the query crop, reproducing the Big-to-Small operating conditions of the UAV-ReID benchmark and directly mapping performance onto engagement distance.
- **Q7:** Q7 — Embedded validation: INT8 quantisation of backbone_tidl.onnx using TIDL calibration, per-crop latency measurement on TDA4VM, and accuracy retention comparison against FP32 baseline.
- **Real data:** Real data incorporation: integrate real fixed-wing UAV footage from the MMFW-UAV dataset (147k images, BIT 2025) and Kaggle fixed-wing collections into the training set, and re-run all experiments to quantify the synthetic-to-real transfer gap.

8.0 Research Questions Status

The following table summarises the status of all seven research questions at the conclusion of the baseline study:

Question	Topic	Status	Key finding
Q1	Enrollment size	Done	5 images recommended for deployment (cross-domain AUC > 0.90 at $k=5$ with real data). Confirmed stable across dataset iterations.

Q2	Enrollment content (which views)	Pending	Not yet evaluated. Planned: compare angle coverage strategies.
Q3	Enrollment degradation	Done	degrade_p=0.5 reduces train/val gap from 0.295 to 0.175. Correct default confirmed.
Q4	Prototype aggregation	Done	Mean beats attention at k≥5 in all conditions. Attention raises impostors proportionally more than genuines.
Q5	Look-alike discrimination	Pending	False-positive characterisation against visually similar impostors not yet run.
Q6	Robustness by distance/pixel size	Pending	Performance stratified by pixel size bucket not yet measured.
Q7	Embedded INT8 on TDA4VM	Pending	TIDL quantisation and on-target latency measurement deferred to Phase 3.

8.1 Prioritised Next Actions

Based on the completed Q1 and Q3 results, the following actions are prioritised in order of expected impact on ROC-AUC:

- **1.** Validate mixed-domain training (immediate). The cross-domain eval_openset result (currently 0.481) is the primary go/no-go metric for the system. If mixed-domain training brings this above 0.80, the deployment problem is solved at the synthetic level and the next step is adding real data. If it fails, the enrollment renders need to be made more operationally realistic before retraining.
- **2.** Real data integration: COMPLETED. Nine real fixed-wing identities added (5 video sequences + 4 Ukrainian UAV exhibition sequences). Cross-domain AUC improved from 0.873 to 0.913 (+0.040). Cross-domain gap closed from 0.030 to 0.010. TPR@10% improved from 0.645 to 0.710. This is now the primary reference model.
- **2.** Expand synthetic identity count to 30+. With 16 training identities, the metric has limited contrast to learn from. Each additional Sketchfab model (10 minutes of render time) adds one new identity. The few-shot literature shows consistent gains until approximately 100 training classes.
- **3.** Add texture and colour randomisation to Blender renders. All current renders use the default .glb material. Random HSV tinting of the UAV surface per render (±30% variation) forces the model to ignore colour and learn shape, directly addressing the largest remaining source of synthetic-to-real domain gap.

- **4.** Swap backbone to EfficientNet-B0 (one line change in model.py). EfficientNet-B0 has similar embedded cost to MobileNetV3-Small but typically achieves 2-3% higher embedding quality on fine-grained recognition tasks. TIDL-compatible with the same split-export strategy.
- **5.** Extend training to 50 epochs with 200 episodes per epoch. Loss was still decreasing at epoch 30. Cosine annealing restarts or longer schedules may yield further improvement at no data cost.
- **6.** Re-run Q4 (attention vs mean) and Q1 (enrollment size sweep) after real data is added. Both results are expected to change: attention is predicted to outperform mean when the gallery contains real client photographs with realistic appearance variation, and the optimal k_{shot} may shift.

9. Preliminary Conclusions

The baseline study establishes three conclusions that directly inform the KAI Vision project.

First, the prototypical network approach is viable for privacy-preserving few-shot UAV identification. A model trained episodically on 16 synthetic UAV identities achieves ROC-AUC = 0.83 on six held-out identities, demonstrating meaningful generalisation to unseen airframes without any client data and without any retraining at deployment time. The go/no-go decision for Phase 2 is positive.

Second, the optimal client enrollment size from the current dataset is 10 images, with 5 as the acceptable minimum. The marginal value of additional images beyond 10 is small. This provides the first empirical basis for the enrollment specification that Embention will present to clients.

Third, the operational degradation pipeline ($\text{degrade}_p=0.5$) consistently outperforms training without degradation across all metrics: ROC-AUC 0.861 vs 0.854, TPR@10% 0.494 vs 0.433, and a substantially reduced train/val gap (0.175 vs 0.295). Degradation acts as domain regularisation, forcing the encoder to learn shape-based features robust to resolution loss rather than render-specific texture features. It is the correct default configuration.

Fourth, there is a clear sweet spot for augmentation severity. Too little augmentation causes overfitting to synthetic rendering artefacts. Too much augmentation — as demonstrated by the aggressive realistic augmentation experiment — destroys the discriminative silhouette information the model needs to tell aircraft apart, causing embedding space collapse (impostor mean score nearly doubling from 0.160 to 0.303, AUC dropping from 0.861 to 0.734). The optimal augmentation level is the maximum that still preserves a recognisable UAV silhouette in every training image. A severity budget parameter should be implemented in the render pipeline before the next dataset generation iteration.

Fifth, the cross-domain matching problem — clean enrollment photographs against degraded operational crops producing random AUC of 0.481 — is fully resolved by mixed-domain episodic training, which draws support images from the enrollment split and query images from the operational split in every training episode. After 50 epochs of mixed-domain training the cross-domain AUC reached 0.873 at $k=10$, while same-domain AUC simultaneously improved to 0.903 — the highest value in the study. The gap between lab performance and deployment performance collapsed from 0.396 to 0.030. This is the primary result of the baseline study.

Sixth, mixed-domain training produced negative impostor mean scores in same-domain evaluation — a qualitatively stronger form of discrimination than any previous model achieved. Where previous models placed impostor aircraft at scores between 0 and 0.16, the mixed-domain model places them at scores between -0.03 and -0.004, on the opposite side of the embedding sphere from the enrolled prototype. This reorganisation of embedding space explains the jump in $\text{TPR}@10\%$ from 0.567 to 0.703.

Seventh, mean prototype aggregation definitively outperforms attention-weighted aggregation at all $k \geq 5$, in both same-domain and cross-domain evaluation, on the final mixed-domain model. Attention raises genuine match scores but raises impostor scores proportionally more, reducing the genuine-impostor gap from 0.708 to 0.493 at $k=10$. Mean aggregation is the correct and final recommendation for the KAI Vision enrollment pipeline.

Eighth, integrating real-world fixed-wing UAV footage into training produced the best results of the entire study. Adding 9 real identities (out of 39 total training identities) pushed cross-domain AUC from 0.873 to 0.913 — the first time this metric exceeded 0.90. The cross-domain gap closed from 0.030 to 0.010, and $\text{TPR}@10\%$ in the deployment scenario improved from 0.645 to 0.710. Real imagery provides appearance variation that synthetic renders cannot replicate and directly closes the synthetic-to-real transfer gap. The real-data mixed-domain model is the final reference model of this baseline study.

Sixth, mean prototype aggregation outperforms attention-weighted aggregation under purely synthetic conditions, but this result is expected to reverse under realistic enrollment conditions. The attention mechanism should be retained in the architecture and re-evaluated in Phase 2 with real client-style photographs.

The primary limitation of this study — exclusive reliance on Blender renders with limited domain randomisation — is the principal driver of the train/val gap and the low TPR at strict operating thresholds. Incorporating real fixed-wing UAV footage and adding texture randomisation to the rendering pipeline are the highest-priority next actions.

10. Research Session Summary — 16 June 2026

This section documents the advances made during the research session of 16 June 2026, including experiments run, discoveries made, and decisions taken.

10.1 What Was Done

The session began with the best model available at the time: the colour mild clean checkpoint trained on 36 identities with mild built-in render degradation, which had achieved a same-domain ROC-AUC of 0.879 and a near-zero train/val gap of 0.024 — the cleanest training result of the study. The first action was to run the cross-domain evaluation that had been identified as the correct deployment metric: enrollment images as gallery, degraded operational val images as queries.

This evaluation returned a ROC-AUC of 0.481 — essentially random. Both genuine and impostor scores were negative. This was not expected from a model that achieved 0.879 in same-domain evaluation, and it forced a careful diagnosis of what had gone wrong.

The diagnosis identified a cross-domain few-shot learning problem: the model had been trained on episodes where support and query images were always drawn from the same image pool, so it had learned to compare operational crops against operational crops. It had never been trained on the actual deployment task of comparing a clean client photograph against a small distant flight crop. The two types of image were being mapped to different regions of the embedding sphere, and cross-region comparison produced meaningless scores.

The solution implemented was mixed-domain episodic training: modifying the episodic sampler so that support images (which form the prototype) are drawn from the enrollment split — clean close-range renders — while query images are drawn from the operational train split — degraded far-away renders. This makes every training episode structurally identical to the deployment task. A `--support_split` argument was added to the training script, and a check was added to restrict episodes to identities present in both splits simultaneously (positive UAV identities only; hard negatives have no enrollment images).

Mixed-domain training ran for 50 epochs on 200 episodes per epoch. The extended training finished with a best val accuracy of 0.645. Full evaluation was then run across both evaluation modes (same-domain and cross-domain) and all k values from 1 to 30.

10.2 What Was Discovered

The results of the mixed-domain model were the most significant of the entire study. Four findings emerged:

- **Cross-domain resolution:** The cross-domain problem was fully resolved. Cross-domain ROC-AUC recovered from 0.481 to 0.873 at k=10. Enrollment photographs and operational crops of the same aircraft are now placed close together in embedding space, and enrollment photographs of different aircraft are placed near zero or below zero. The gap between same-domain performance and cross-domain performance collapsed from 0.396 to 0.030.
- **Simultaneous improvement:** Same-domain performance simultaneously improved. The mixed-domain model achieved a same-domain ROC-AUC of 0.903 at k=10 — the first result in the study to break 0.90 — despite training on only 20 shared identities rather than the 36 used by the previous best model. Mixed-domain training did not sacrifice same-domain performance to gain cross-domain performance; it improved both at the same time.
- **Negative impostor scores:** The impostor mean went negative for the first time. All previous models placed impostor aircraft at cosine similarity scores between 0.06 and 0.16. The mixed-domain model places them at -0.029 to -0.004 in same-domain evaluation — on the opposite side of the embedding sphere from the enrolled prototype. This qualitatively stronger form of discrimination explains the jump in TPR@10% to 0.703 and suggests that learning a quality-invariant representation forces the model to encode shape in a more abstract, orthogonal way.
- **Q1 revised:** The Q1 fine-grained sweep revised the enrollment recommendation. Running the k-shot sweep at every integer from 1 to 30 showed that the plateau begins at k=3, not k=10 as suggested by the earlier coarse sweep. The k=10 recommendation was an artefact of high sampling variance with only 11 val identities and a small number of k values tested. The definitive recommendation is minimum 3 enrollment images, 5 as a comfortable safe choice, beyond 5 gives no reliable benefit. This reduces the client burden compared to the previous specification.

A fifth finding confirmed a pattern observed in earlier experiments: mean prototype aggregation definitively outperforms attention-weighted aggregation across the full sweep from k=2 to k=30 and in both evaluation modes. The fine-grained sweep made the mechanism completely transparent: attention raises genuine scores strongly as k increases (genuine score rises from 0.402 at k=1 to 0.877 at k=30) but raises impostor scores proportionally more (from -0.037 to +0.421), shrinking the genuine-impostor gap continuously. Mean keeps impostor scores near zero throughout the sweep while genuine scores grow more slowly. The gap for mean at k=20 is 0.662 versus 0.478 for attention. Q4 is definitively closed.

10.3 Decisions Taken

Based on the session results, the following decisions are fixed for the remainder of the project:

- Primary evaluation metric is cross-domain ROC-AUC (enrollment gallery, val queries). Same-domain AUC is reported for literature comparison but is not the deployment metric.
- Aggregation method: mean prototype. Attention is documented as evaluated and rejected, with the mechanism explained.
- Enrollment recommendation: 5 images minimum, the same as before, but the justification is now based on the fine-grained sweep rather than the coarse one. 3 is the statistical knee; 5 is the safe recommendation that accounts for sampling variance.
- Training configuration: mixed-domain episodic training with `--support_split` enrollment is the correct baseline for all future runs. `degrade_p=0.0` because degradation is baked into the renders.
- A real-world demo is feasible now. The pipeline (`enroll.py` + `identify.py`) works end-to-end. Photographing a real RC aircraft from close range for enrollment and from 30-50 metres for operational queries and running the pipeline on those photos is a concrete proof-of-concept deliverable for Embention.

10.4 What Comes Next

The extended training ran for 100 epochs and confirmed convergence: loss plateaued at epoch 80 (1.094), train accuracy reached 0.977 against a val accuracy of 0.606, and the best checkpoint (epoch 22, val acc 0.643) was slightly worse than the 50-epoch model (0.645). More training epochs do not improve results. The 50-epoch mixed-domain checkpoint is the definitive best model of the baseline study.

The real-world deployment demo exposed a pipeline gap: the model was trained on full Blender frames but at runtime receives YOLOX crops — a different input distribution. The appropriate response is retraining with YOLOX-cropped renders as query images, which closes the gap between training format and deployment format. The YOLOX crop dataset has been assembled and audited (`uav_dataset_yolox_crops`, 40 train identities, 14 val identities, 68% of crops in the 46–143px operational envelope). Training is the next immediate action.

Following YOLOX crop training: Q5 (look-alike discrimination), Q6 (performance by pixel size / engagement distance), and Q7 (TIDL INT8 embedded validation on the TDA4VM) remain pending. The end-to-end real-world demo (genuine topk 0.690, impostor topk 0.192 at topk=15) has already validated the pipeline concept and is ready to present to Embention.

11. Final Model Summary

The YOLOX crop mixed-domain model (checkpoints_yolox_crops_mixed_domain/best.pth, epoch 8) is the final reference model of this study. Its defining characteristics:

- Training query format matches deployment: YOLOX-cropped small UAV images against sky, 30–110px crop size
- Enrollment format matches client photos: large close-range renders, >300px, one per distinct viewpoint
- Cross-domain AUC 0.955 at k=5 — highest in the study
- Same-domain AUC 0.962 at k=10 — highest in the study
- Cross-domain gap 0.007 — essentially zero: enrollment photos and operational crops treated identically
- TPR@10% 0.875 at k=5 cross-domain — 87.5% genuine detection rate at 10% false alarm rate
- Privacy architecture validated: 5 client photos → 2.6 KB gallery.npy, original images never transmitted
- Zero-shot generalisation confirmed: YouTube UAV never seen in training, correctly identified at TopK@15 = 0.592 vs impostor 0.369

Pending work before the study is complete: Q5 (look-alike discrimination between visually similar airframes), Q6 (performance stratified by pixel size and engagement distance), and Q7 (TIDL INT8 quantisation and on-target latency validation on the TDA4VM SoC).

12. Deployment Engineering Report

This section documents the end-to-end deployment pipeline as a standalone engineering report, focused on operational behaviour rather than academic metric analysis. It covers what was built, what was measured in real deployment conditions, and what the results mean for a working product.

12.1 Objective

The operational goal is deliberately simple and binary. A client enrolls a single target UAV by providing approximately five reference images. From that point, for every frame the camera sees, the system answers exactly one yes/no question: does the UAV in this frame match the enrolled target? It never ranks the target against a list of known drones. It never retrain for a new target. The training identities are irrelevant at runtime — only the enrolled gallery matters. This binary, gallery-only regime is what a deployed device would actually run in, and all measurements below were taken against that regime.

12.2 What Was Built

An end-to-end pipeline was assembled from existing components plus new glue code. The processing chain is: raw camera frame → YOLOX detector (UAV crop extraction) → ProtoNet encoder (128-dimensional embedding) → cosine similarity against enrolled gallery prototype → threshold decision (MATCH or UNKNOWN) → multi-frame aggregation into a final verdict. This structure mirrors deployment reality: detect, crop, embed, compare to the one enrolled gallery, decide.

The following components were created or hardened during this phase:

src/deploy.py — The Verifier class. Loads the trained model and enrolled gallery, embeds any crop, and produces the match score. Made the single source of truth so calibration and deployment always use identical scoring logic.

src/enroll.py — Converts five reference images into a ~2 KB gallery file (gallery.npy), the only artifact that ever leaves the client's premises. Central to the privacy architecture.

scripts/calibrate_threshold.py — Feeds positive and negative crops through the scorer and recommends a decision threshold from data rather than by guessing. Reports detection rate, false-accept rate, and balanced accuracy.

scripts/deploy_verify.py — The one-command demo. Accepts a video file or folder of frames, runs the full pipeline, writes a per-frame CSV, and prints a verdict. Includes crop caching (re-scoring against a new gallery skips re-extraction), annotated overlays, Grad-CAM heatmaps, and a median-score gate.

scripts/plot_embedding_map.py — Visualises the embedding space through 2D/3D projections and a full 128-dimensional pairwise similarity matrix coloured by identity, so cluster quality can be seen rather than inferred from numbers alone.

The verdict rule is: CONFIRMED if at least 3 frames match AND at least 25% of detected frames clear the threshold AND the median frame score clears the threshold. The median gate was added specifically so an impostor cannot pass by grazing the threshold on a minority of lucky frames — the median describes the typical frame and ignores outliers.

12.3 Experimental Results

12.3.1 Happy Path Validation

The enrolled demo target was correctly confirmed and an impostor video was correctly rejected, proving the full pipeline worked end to end with sensible binary decisions.

12.3.2 Stress Test: Clear vs Look-alike Impostors

Four real UAV videos (skywalker, hornet, vector, lancet) were scored against multiple galleries. The results split sharply:

Test case	True target median	Impostor median	Gap	Result
Clear impostor (lancet vs skywalker)	+0.31	-0.25	0.56	Clean separation
Look-alike (hornet vs skywalker)	+0.41	+0.46	-0.05	Impostor scores higher
Look-alike (vector vs skywalker)	similar	similar	~0	Coin flip (51.8% acc)

Against clearly different aircraft the separation is clean and the verdict is reliable. Against visually similar fixed-wing UAVs at operational crop sizes, the embeddings collapse together and no threshold or aggregation rule can separate what the encoder does not distinguish. This is a representation limit, not a tuning problem.

12.3.3 Metric Correction: Median Replaces TopK Mean

The original demo summary metric reported the average of the top-15 frame scores, which cherry-picks outlier frames and makes impostors appear closer to the target than they typically are. This was replaced with the median frame score and the percentage of frames above threshold — robust numbers that describe the typical frame. This change also exposed a false confirmation case where an impostor gallery was scoring 40% of frames above threshold with a median of only 0.12. The median gate correctly blocked this false positive while keeping the true target confirmed.

12.3.4 Background Contamination Confirmed by Grad-CAM

Grad-CAM heatmaps revealed that the model's attention was concentrated on the sky background rather than the airframe silhouette. This was confirmed by a direct controlled test: removing the background from query crops shifted scores dramatically. The enrolled target jumped from a median of 0.34 to 0.69 after background removal, while a similar-looking impostor dropped from 0.71 to 0.49. With background present, the true target had actually scored lowest of all tested galleries; with background removed, the ranking corrected to the expected order. This is the most direct evidence that background contamination is a primary failure mode.

12.3.5 Background Masking Attempt

A colour-based background mask was applied at training time to test whether heuristic background removal could improve the encoder. The mask degraded small UAV crops by removing pixels from the aircraft itself at sizes below approximately 50px, and the approach was reverted. The conclusion: a clean background removal fix requires pixel-perfect Blender alpha masks during rendering, not a heuristic applied to compressed JPEG crops where the aircraft and sky share similar pixel values at small sizes.

12.3.6 Retraining with Real Data

A new model was trained with real-world images added to the dataset (42 training identities, 15 validation identities, mixed-domain enrollment support, euclidean metric). Best validation accuracy was 0.811 with training accuracy at 1.000, a gap of approximately 19 percentage points. This gap signals that the model memorises the specific appearance of training identities rather than learning a fully general metric. Real-world generalisation sits at approximately 0.81 — reliable against clearly distinct UAVs, not yet reliable against look-alikes at operational resolution.

12.4 Interpretation

The pipeline is sound and demonstrable. Enrolling once and verifying per-frame against only the enrolled gallery — the correct deployment regime — produces clean binary verdicts when the impostors are clearly different from the target. Aggregating across many frames makes the system robust to occasional bad frames.

The encoder partially identifies sky rather than airframe. This confound inflates impostor scores whenever the sky conditions match and is the single most cost-effective weakness to address, since it is a data and rendering issue rather than an architectural one.

The encoder cannot distinguish UAVs that genuinely look alike at 30–100px crop sizes. Their embeddings collapse together in the 128-dimensional space. No threshold, gate, or aggregation strategy can separate what the embeddings do not encode — this is a fundamental representation limit at operational resolution, not a tuning problem.

12.5 Prioritised Next Steps

The following improvements are ordered by expected impact on the look-alike discrimination failure, which is the primary operational limitation:

- 1. Hard-negative training:** Teach the encoder using visually similar UAVs as explicit hard negatives within episodes, so it learns the fine structural differences that distinguish look-alike airframes. Highest leverage on the core failure mode.
- 2. Background handling:** Keep enrollment images background-removed. Apply size-conditional background removal to query crops above approximately 80px while skipping tiny crops where masking degrades quality. Generate clean masks via Blender

alpha channel at render time rather than applying heuristics to compressed crops post-hoc.

3. More real frames per identity: Natural background variety across real footage teaches background invariance. Never duplicate the same crop with and without background as separate training examples — that is memorisation of the processing artifact, not learning.

4. Reduce overfitting: Maintain the degrade augmentation pipeline, deploy best.pth rather than last.pth, and consider adding dropout or weight decay to the projection head.

5. Resolution gate: For crops below approximately 40px, report DETECTED but IDENTITY-UNCERTAIN rather than a confident match or rejection verdict, since identity information is not recoverable at that resolution.

Single highest-impact action: fine-tune the encoder on real UAV crops with hard negatives from visually similar aircraft.